1 Relation to Solomonoff operator induction

Solomonoff operator induction, the subject of the companion paper (Hänni and Claude (Anthropic), 2026b), puts the universal prior on programs that label inputs. On sentences, a labeller consistent with B is the program-shaped counterpart of a theory, and we compare SAI with SOI over such labellers.[1] Table 1 collects the relations, and Appendix A has the proofs. Losses are in the semimeasure convention unless stated; $\asymp$ is equality within $2^{\pm c}$ on every finite data set.

1.1 The objects and four maps

Definition 1.1 (The operator-induction side)**.** Let $\mathcal{L}$ and B be as in **??**.

- A *function machine* is a c.e. set $F \subseteq \mathrm{Sent} \times \{0,1\}^* \times \{0,1\}$ such that $(\varphi, w, y), (\varphi, w', y') \in F$ with w, w' comparable implies $y = y'$ (Definition 3.6 of the companion paper). On a tape ω it computes the partial labeller f^F_ω: $f^F_\omega(\varphi) := y$ if $(\varphi, w, y) \in F$ for a prefix w of ω, and $f^F_\omega(\varphi)$ is undefined ($\perp$) otherwise; its law is $\nu_F(D) := \mathbb{U}\{\omega : f^F_\omega \supseteq D\}$. F_U is the universal function machine (read $p \in \mathrm{dom}\, V$, then run $F_{V(p)}$ on the rest of the tape), $f_\omega := f^{F_U}_\omega$, and $M_F := \nu_{F_U}$.
- For a partial $f : \mathrm{Sent} \rightharpoonup \{0,1\}$ let $\Gamma_f := \{\varphi^{f(\varphi)} : \varphi \in \mathrm{dom}\, f\}$. The labeller f is *consistent* if $B \cup \Gamma_f$ is consistent, and *closed* if moreover $f(\varphi) = b$ iff $B \cup \Gamma_f \vdash \varphi^b$, for all φ and b.
- *One-tape consistent and closed SOI*: $M^{\mathrm{cons}}_F(D) := \mathbb{U}\{\omega : f_\omega \supseteq D,\ f_\omega \text{ consistent}\}$, and $M^{\mathrm{cl}}_F(D)$ likewise with "closed and consistent"; ν^{cons}_F, ν^{cl}_F are the same for one machine F.
- *Random deciders* are function machines F with f^F_ω total for every ω, read as theory machines with axiom set $A^{\mathrm{dec}}_F(\omega) := \{\chi : f^F_\omega(\chi) = 1\}$; $M^{\mathrm{dec}}_A(D)$ is the sum of $2^{-|p|}\,\mathbb{U}\{\omega : B \cup A^{\mathrm{dec}}_{F_{V(p)}}(\omega) \text{ consistent and } \vdash \psi_D\}$ over the $p \in \mathrm{dom}\, V$ for which $F_{V(p)}$ is a random decider.
- *Coin-free and Kolmogorov*: $W^{\mathrm{cons}}(D)$ is the sum of $2^{-|p|}$ over the $p \in \mathrm{dom}\, V$ for which the partial computable labeller $\{V(p)\}$ is consistent and extends D, and $K^{\mathrm{cons}}_F(D)$ is the least such $|p|$; W^{cl}, K^{cl}_F likewise with closed labellers; W^{all} drops the consistency condition (plain coin-free SOI on sentences). W and K_F are the companion paper's coin-free and Kolmogorov versions on strings.

Four maps. The equivalences rest on four computable maps, each a constant-size prefix in front of a program (Lemma A.2): *proof search* turns a theory machine into a labeller that, where the theory is consistent, is closed and decides every sentence as the theory does; *emission* turns a labeller f into the theory Γ_f; *Craig's map* turns f into a random decider of padded copies of Γ_f; *closure* makes a consistent labeller closed.

1.2 Equivalence, version by version

Theorem 1.2 (Coin-free equivalence)**.** *There is c, depending only on V, the coding of syntax and B, such that for every finite data set D:*

(a) $2^{-c}W_A(D) \le W^{\mathrm{cons}}(D) \le 2^c W_A(D)$, $2^{-c}W_A(D) \le W^{\mathrm{dec}}_A(D) \le 2^c W_A(D)$, *and* $2^{-c}W^{\mathrm{cons}}(D) \le W^{\mathrm{cl}}(D) \le W^{\mathrm{cons}}(D)$*;*

(b) $K_A(D)$, $K^{\mathrm{dec}}_A(D)$, $K^{\mathrm{cons}}_F(D)$ *and* $K^{\mathrm{cl}}_F(D)$ *agree up to $\pm c$, and all are finite iff D is consistent;*

[1] The comparison of SAI with consistent SOI, the example in which plain SOI beats SAI (Remark A.14), and the axiom schema of Remark A.3 first appeared in Hänni's working notes (Hänni, 2026).

(c) for every $\varphi \notin \operatorname{dom} D$ and $o \in \{1, 0, \bot\}$, $2^{-c} W_A(D; \varphi \mapsto o) \le W^{\mathrm{cl}}(D; \varphi \mapsto o) \le 2^c W_A(D; \varphi \mapsto o)$, where $W_A(D; \varphi \mapsto o)$ is the weight of the enumerators compatible with D whose theory proves φ ($o = 1$), proves $\neg\varphi$ ($o = 0$) or decides neither ($o = \bot$), and $W^{\mathrm{cl}}(D; \varphi \mapsto o)$ is the weight of the closed labellers extending D with $f(\varphi) = o$.

Hence the semimeasure losses of W_A, W_A^{dec}, W^{cons} and W^{cl} agree within $2c$ on every data set, in every order.

Proof search and emission give $W_A \asymp W^{\mathrm{cons}}$; Craig's map is needed only for deciders, and for them (a) is Theorem 6.3 of our earlier report (Hänni and Claude (Anthropic), 2026a). The construction is Craig's; apart from that report we have not found the weight equivalence stated in the literature. A route through an axiom schema works with two sorts or with bounded witnesses but fails in one sort over PA (Remark A.3).

Theorem 1.3 (One-tape equivalence)**.** *There is c, depending only on V, the coding and B, such that for every finite D, the empty one included:*

(a) $2^{-c} M_A(D) \le M_F^{\mathrm{cons}}(D) \le 2^c M_A(D)$ and $2^{-c} M_A(D) \le M_A^{\mathrm{dec}}(D) \le 2^c M_A(D)$;

(b) tape by tape: for every theory machine G and tape ω, if $B \cup A_\omega^G$ is consistent and proves ψ_D, then $f_\omega^{F_G}$ is consistent and extends D; for every function machine F and tape ω, if f_ω^F is consistent and extends D, then G_F and Craig's random decider of F are compatible with D on ω.

Over closed labellers the three outcome weights agree as well, so the trichotomy of SAI is the three-way split of one-tape SOI over closed labellers (Corollary A.4); over all consistent labellers it is not (Proposition 1.5).

Penalties on the axiom side. Charging a hypothesis for the time it takes to check or to generate its axioms, with deduction free ($M_A^{\mathrm{chk},t}$, $M_A^{\mathrm{gen},t}$ and the coin-free $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ of Definition A.7), at any time-constructible budget $t(m) \ge m^{a^*}$ (a^* from Assumption A.5), changes nothing up to constant factors, also class by class: one-tape under an assumption ($\mathrm{S}_{\mathrm{tape}}$) of efficient self-simulation of the universal machine, coin-free under its tape-free form (S), neither of which we prove for a particular machine (Assumptions A.5 and A.6, Theorem A.9, and Propositions A.10 and A.28).

1.3 Positive literals: SAI is SOI

Theorem 1.4 (Transfer principle)**.** *Let $\mathcal{I} \subseteq \{0,1\}^*$ be the companion paper's (infinite, decidable) input set and $x \mapsto t_x$ a computable injective map from $\mathcal{I}$ to closed $\mathcal{L}$-terms, with decidable range and computable inverse. Let R be a unary relation symbol that does not occur in B, and suppose some model of B interprets the terms t_x as pairwise distinct elements. For D with $\operatorname{dom} D \subseteq \{R(t_x) : x \in \mathcal{I}\}$ put $D^{\mathrm{str}} := \{(x, b) : (R(t_x), b) \in D\}$. Then there is c with*

$$|K_A(D) - K_F(D^{\mathrm{str}})| \le c, \qquad W_A(D) \asymp W(D^{\mathrm{str}}), \qquad M_A(D) \asymp M_F(D^{\mathrm{str}}) \quad (D \neq \emptyset),$$

where K_F, W, M_F are the companion paper's objects over $\mathcal{I}$ with labels $\{0,1\}$; the same holds for K_A^{dec}, W_A^{dec}, M_A^{dec}, and for the three outcome weights of a literal query $R(t_y)$ (proves, refutes, decides neither; label 1, 0, undefined at y).

pair	relation (hypotheses)	where
Equivalences		
W_A, W_A^{dec} – $W^{\mathrm{cons}}, W^{\mathrm{cl}}$	$\asymp$, also the outcome weights against W^{cl}; K_A, K_A^{dec}, K_F^{cons}, K_F^{cl} agree $\pm c$, finite iff D consistent	Thm. 1.2
M_A, M_A^{dec} – $M_F^{\mathrm{cons}}, M_F^{\mathrm{cl}}$	$\asymp$, tape by tape; outcome weights against M_F^{cl}	Thm. 1.3, Cor. A.4
$M_A^{\mathrm{chk},t}, M_A^{\mathrm{gen},t}$, $W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}$	$\asymp$ unpenalised, also class by class ($t \geq m^{a^*}$; ($\mathrm{S_{tape}}$), (S))	Thm. A.9, Prop. A.10
$W_A^{\mathrm{Kt},t}$, under (S)	between $2^{-c}W^1$ and $2^c t(m_0)W^1$; a labeller pays $\log(\lvert f\rvert+1)+O(1)$	Prop. A.23
positive literals – strings	$\asymp$, $\pm c$, version by version (fresh predicate, distinct terms)	Thm. 1.4, Cor. A.12
all – observational axioms	$\asymp$ in every version, class by class (free deduction)	**??**
r.e. in $\mathcal{L}$ – finite in $\mathcal{L}^+$	$K_{\mathrm{fin}} = K_A \pm c$, $W_{\mathrm{fin}} \asymp W_A$, $\mathcal{L}^+$ three-sorted; not one-tape ($\mathcal{L}$ finite)	**??**
charged SAI– certificate SOI	size, material: $\asymp$ up to $O(1)$ bits and polynomial budget changes	Thm. A.31
finite axiom sets – circuits	verifier circuits: polynomially related; Bayes over them (literal data)	Thm. A.43, Cor. A.44
Gaps between versions		
2^{-K_A} – M_A, two atoms	no coding theorem: $K_A \geq \log d - 2 - c$, $-\log M_A \leq 2 + c$	**??**, Cor. A.12
W_A – M_A, literals, $\lvert w\rvert \leq d$	gap $\leq \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c$, δ_D the labels' randomness deficiency	Prop. A.13
W^{cons} – M_F^{cons}, W_A – M_A	on the lottery M_F^{cons} is bounded, W^{cons} unbounded, W^{all} bounded	**??**
M_A – W_A on Π_1 data	a coding theorem holds (B true, $\supseteq \mathsf{Q}$)	**??**
M_F^{cons} – machine-wise M_F^{mw}	$\asymp$ if finite consistency is decidable; open in general	Prop. A.21, Q. A.22
computability	none lsc if finite consistency is not r.e.; $\emptyset'$ resp. $\emptyset''$	Prop. A.19
Separations		
SAI– W^{all}	W^{all} wins by n bits on coin literals labelled 1 (encoding effect)	Rem. A.14
entailed query	SAI certain; consistent SOI keeps undefined mass $\geq 2^{-K(\chi)-c}$	Prop. 1.5
observed ι, $\mathsf{Q} \subseteq \mathrm{Cn}(B)$	one-sided; gap unbounded on one datum, bounded on r.e. families	Prop. 1.6, Prop. A.17
$W^{\mathrm{cons},\tau}$, $M_F^{\mathrm{cons},\tau}$; B true	regret $\leq c_\tau$; reverse $\geq n - 1.37 - c_\perp$ on a computable sequence	Cor. A.25, Thm. 1.7
SAI– $W^{\mathrm{all},\tau}$	incomparable	Prop. A.26
W^{cons}, M_F^{cons}, diagonal data	unpenalised loss $\leq \lvert u\rvert + c$: the gap is free deduction	Prop. 1.8, Thm. A.29
clocked SOI, Presburger	SAI bounded; clocked unbounded for $\tau \leq 2^{2^{c'm}}$ (coin-free), $\tau \leq 2^{c'm}$ (one-tape); doubly exponential one-tape conditional	Prop. A.37, Prop. A.40
charged SAI, Presburger	size, material: doubly exp.; (a1): exp., doubly exp. open; (a2), (b2) open	Prop. A.41, Q. A.42
Fit classes under charged deduction (literal data, polynomial budgets)		
readings (b); (a1); (a2), (b2)	$\mathsf{NP} \cap \mathsf{coNP}$; $\mathsf{EXPTIME}$; every decidable set	Cor. A.32
observational, run-padded, finite	see **??**; finite theories need a binary relation or entities ($B = \emptyset$)	**??**, **??**

Table 1: SAI and SOI. Semimeasure convention unless stated; $\asymp$ is equality within $2^{\pm c}$ on every finite D; B is decidable and consistent; rows with Q or "B true" assume $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$; charged rows assume (F1); only "open" and "conditional" entries are not theorems.

Distinct terms are needed: if $B \vdash t_x = t_y$, the data $(R(t_x), 1), (R(t_y), 0)$ are inconsistent for SAI but not for SOI.

So every result of the companion paper on binary labels that holds up to additive constants in K, $-\log W$ and $-\log M_F$ holds for SAI on literal data, among them the absence of a coding theorem with its two-point rate; normalised cumulative statements, computability statements, and statements about private randomness or weak semimeasures do not transfer in general (Corollary A.12). The content of SAI beyond SOI lies in logically structured data. Plain coin-free SOI can beat SAI by n bits in expectation on literals with random signs, all labelled true, through the inconsistent labeller "always 1": an encoding effect (Remark A.14).

1.4 What breaks

Proposition 1.5 (Logical closure)**.** *Let D be consistent and $\chi \notin \operatorname{dom} D$.*

(a) If $B \cup D^{\pm} \vdash \chi$, every hypothesis of SAI compatible with D proves χ: $M_A(1 \mid D, \chi) = 1$, $M_A(0 \mid D, \chi) = 0$ and $\mathrm{Indep}_D(\chi) = 0$; the same for W_A.

(b) If $B \cup D^{\pm} \vdash \chi$, then $W^{\mathrm{cons}}(0 \mid D, \chi) = 0$ and $W^{\mathrm{cons}}(1 \mid D, \chi) \geq 2^{-2c}$; the same for M_F^{cons}.

(c) For every $\chi \notin \operatorname{dom} D$, entailed or not, the consistent labellers extending D that are undefined at χ have weight at least $2^{-K(\chi)-c}\, W^{\mathrm{cons}}(D)$, and the tapes on which f_ω is consistent, extends D and is undefined at χ have measure at least $2^{-K(\chi)-c}\, M_F^{\mathrm{cons}}(D)$.

(d) Over closed labellers, the three predictive probabilities agree with those of SAI within $2^{\pm 2c}$.

So in the semimeasure convention consistent SOI never becomes certain of a logical consequence: its undefined mass on a fixed simple χ stays above $2^{-K(\chi)-c}$ whatever the data (in the normalised convention its prediction on an entailed χ is 1, by (b)). The equivalences control the losses, not the undefined mass, which mixes independence with laziness.

An observed "independent" label. A labeller f with values in Y_3 is *consistent* if $B \cup \Gamma_f$ is consistent (Γ_f formed from the labels 1, 0) and decides none of the sentences f labels ι. $W_A^3(D)$ is the weight of the enumerators whose theory is consistent, proves the 1/0 data of D and decides no ι-labelled sentence of D; $W^{3,\mathrm{cons}}(D)$ is the weight of the consistent three-valued labellers extending D; M_A^3 is as in **??**, and $M_F^{3,\mathrm{cons}}$ is M_F^{cons} with labels in Y_3. Ind_B is the set of sentences that B decides neither way.

Proposition 1.6 (An observed independent label breaks the equivalence)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$.*

(a) $2^{-c} W^{3,\mathrm{cons}}(D) \leq W_A^3(D)$ for every three-valued D, and $M_F^{3,\mathrm{cons}}(D) \leq 2^c M_A^3(D)$ for every three-valued $D \neq \emptyset$.

(b) For every n there is $\sigma_n \in \mathrm{Ind}_B$ with $W^{3,\mathrm{cons}}(\{(\sigma_n, \iota)\}) \leq \Omega_V^{\geq n+1}$ while $W_A^3(\{(\sigma_n, \iota)\}) \geq 2^{-c}$; and there is $\sigma'_n \in \mathrm{Ind}_B$ with $M_F^{3,\mathrm{cons}}(\{(\sigma'_n, \iota)\}) \leq 2^{1-n} + \Omega_V^{\geq n+1}$ while $M_A^3(\{(\sigma'_n, \iota)\}) \geq 2^{-c}$.

(c) For every $\sigma \in \mathrm{Ind}_B$, $W_A^3(\{(\sigma, \iota)\}) \leq 2^{K(\sigma)+c}\, W^{3,\mathrm{cons}}(\{(\sigma, \iota)\})$ and $M_F^{3,\mathrm{cons}}(\{(\sigma, \iota)\}) \geq 2^{-K(\sigma)-c}$.

The mechanism: a theory leaves σ undecided, a Π_1 property, without committing to anything, while a labeller must output ι after finitely many steps. The sentences that short consistent labellers label ι form an r.e. subset of the productive set Ind_B (Lemma A.16), while an r.e. family of independent sentences is labelled ι by one consistent labeller, so no such family shows a large gap (Proposition A.17).

Consistency constraints of unbounded arity separate the versions of consistent SOI (**??**). None of the versions is lower semicomputable when consistency of finite data sets is not r.e. (Proposition A.19).

1.5 Time bounds with free deduction

On the labeller side a labeller runs for a bounded time: for a time-constructible τ and a string f, let $f^\tau(\varphi) := U_0(f, \varphi)$ if a fixed plain universal machine U_0 halts within $\tau(|\varphi|)$ steps with a label, and let f^τ abstain otherwise; f has weight $w(f) := 2^{-|E(f)|}$ for a prefix code E whose weights sum to 1 with computable tails. *Clocked consistent* SOI is $W^{\mathrm{cons},\tau}(D) := \sum_f w(f)\mathbf{1}[f^\tau$ consistent and $\supseteq D]$; $W^{\mathrm{all},\tau}$ drops consistency, and $M_F^{\mathrm{cons},\tau}$, $M_F^{\mathrm{all},\tau}$ are the one-tape forms (Theorem A.29). Let $c_\perp := -\log w(f_\perp)$ for the always-abstaining $f_\perp$. A *computable predictor* P maps D and $\varphi \notin \operatorname{dom} D$ to a pair (q_0, q_1), $q_0 + q_1 \leq 1$, computable to any precision; its loss is $\ell_P(D_n) := \sum_{j \leq n} -\log q_{b_j}(D_{j-1}, \varphi_j)$.

On every data set the losses of W_A and W_A^{dec} exceed that of $W^{\mathrm{cons},\tau}$, and the loss of M_A that of $M_F^{\mathrm{cons},\tau}$, by at most a constant c_τ; under (S) the same holds for the penalised versions (Corollary A.25). The reverse fails on sentences that encode a diagonal argument. Let $\mathrm{Out}(e,x,y)$ be a Σ_1 formula without $<$ that holds in $\mathbb{N}$ iff program e halts on input x with output y (by the MRDP theorem; Lemma E.14 of the earlier report), and let DET be the true Π_1 sentence saying that no program has two outputs on one input.

Theorem 1.7 (Diagonal separation)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and let B be decidable and true in an expansion of $\mathbb{N}$. For every computable predictor P there is a program e such that, with $s_n := \mathrm{Out}(\underline{e},\underline{n},S0)$ labelled by truth: e is total with values in $\{0,1\}$; $\mathsf{Q} \cup \{\mathrm{DET}\}$ decides every s_n; $\ell_P(D_n) \geq n - \sum_j \log(1+2^{-j-2}) \geq n - 1/(4\ln 2) \geq n - 0.361$; and W_A, $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ lose at most $\log(1/\omega_0)$, where $\omega_0 > 0$ is the least of the weights of a fixed decider and a fixed enumerator of $\mathsf{Q} \cup \{\mathrm{DET}\}$ in these mixtures, independent of P. Moreover, on the sequence built in this way against the computable predictor $P^*(D) := (W^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$, $W^{\mathrm{all},\tau}$ loses at least $n - 1.37$ and $W^{\mathrm{cons},\tau}$ at least $n - 1.37 - c_\perp$.*

One-tape, $M_F^{\mathrm{all},\tau}$ and $M_F^{\mathrm{cons},\tau}$ lose at least $n-1.37$ and $n-1.37-c_\perp$ on a computable sequence on which M_A loses at most $\log(1/\omega_0)$ (Theorem A.29).

Clocked plain $W^{\mathrm{all},\tau}$, which may be inconsistent, and SAI are incomparable (Proposition A.26): SAI is not stronger than every time-bounded predictor, only strictly stronger than clocked *consistent* SOI. Nor is this gain one of axioms over programs:

Proposition 1.8 (The separation comes from free deduction)**.** *(a) Let u be the labeller that, on a sentence $\mathrm{Out}(\underline{e},\underline{n},S0)$, runs e on n and outputs 1 if the result is 1 and 0 if it is anything else (diverging if e diverges), and is undefined on all other sentences. Then $\Gamma_u \subseteq \mathrm{Th}(\mathbb{N})$, so u is consistent with every B true in an expansion of $\mathbb{N}$, and $|u|$ is a constant. (b) On every sequence of Theorem 1.7,* unpenalised W^{cons} *(and W^{all}) has loss at most $|u| + c$.*

What the time bounds show. Penalising the axiom side changes nothing, even class by class; penalising the labeller side makes the relation one-sided, and the gap is the free deduction of SAI, which unpenalised consistent SOI has too. So we read the time-bounded separations as showing not that axioms are better hypotheses than programs, but that free deduction is free computation. (The defeating sequence is computable within a slightly larger time bound; we only sketch this, Remark A.27.)

1.6 Charged deduction

The natural repair is to charge deduction, in the setting of Definition E.18 of the earlier report (Appendix A.6): a hypothesis decides its axiom set in bounded time, and each datum needs a derivation of bounded size, or one in which every axiom instance (reading (a1)) or every nonlogical instance ((a2)) is short, or one of bounded material, the total size of the distinct instances used ((b)), or of bounded nonlogical material ((b2)). Under the renaming condition (F1) of that report (Assumption A.30) and with time-constructible nondecreasing budgets, derivation-charged SAI is certificate SOI (labellers defined by time-bounded verifiers of short certificates), up to $O(1)$ bits and polynomial changes of the budgets (Theorem A.31): a derivation is a certificate. On literal sequences at polynomial budgets with charged degrees this fixes the fit classes: $\mathsf{NP} \cap \mathsf{coNP}$ under (b), $\mathsf{EXPTIME}$ under (a1), every decidable set under (a2) and (b2) (Corollary A.32).

What is charged. Every reading bounds the size of a derivation that *exists* and none charges *finding* it, so charged deduction, of the type $\mathsf{NP} \cap \mathsf{coNP}$, cannot express the point that a written theory is far from sufficient for a simple reasoner; a deterministic bounded reasoner can (**??**).

1.7 A natural separation: Presburger arithmetic

A natural theory separates as well. Let $\mathcal{L}_{\mathrm{Pr}} := \{0, 1, +\}$ (with equality), $B = \emptyset$, $\mathrm{PrA} := \mathrm{Th}(\mathbb{N}, 0, 1, +)$, and let D^{Pres} list all $\mathcal{L}_{\mathrm{Pr}}$-sentences in length-lexicographic order, labelled by truth. By the bound of Fischer and Rabin (1974), under a proviso on their measure of length, there is $c > 0$ such that for all large m every decision procedure for PrA, deterministic or nondeterministic, takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols (Remark A.36).

On D^{Pres}, W_A, W_A^{dec}, M_A, and $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$, $M_A^{\mathrm{chk},t}$ for a large enough polynomial t, have bounded loss through the decider of Presburger's axioms (Presburger, 1929); given the Fischer–Rabin bound, the losses of $W^{\mathrm{cons},\tau}$ and $W^{\mathrm{all},\tau}$ tend to infinity for clocks $\tau(m) \le 2^{2^{c'm}}$ (m large, $c' < c$), and those of their one-tape forms for $\tau(m) \le 2^{c'm}$ (Proposition A.37). Under a Σ_2-time lower bound for PrA and a time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$, the one-tape losses are unbounded for doubly exponential clocks as well (Proposition A.40).

With deduction charged, SAI needs doubly exponential budgets on these data under the size and material readings, and at least exponential instance sizes under (a1); whether (a1) needs doubly exponential ones, and what (a2) and (b2) need, is open (Proposition A.41 and Question A.42). So on a natural theory, as on the diagonal sequences, the separation from clocked SOI is the free deduction, for the readings in which the question is settled (Appendix A.7).

1.8 Summary

How to read the table. The equivalences hold version by version, also with the axiom side penalised; charged deduction is certificate SOI. On literals SAI and SOI coincide; the separations we know come from logical closure, observed independence, consistency constraints of unbounded arity, or deduction that is free. In the idealised setting, on data labelled true or false, every consistent labeller is an axiom system at a cost of $O(1)$ bits and every theory a closed labeller, so the formalism cannot see the difference in shape between a theory and a program; **????** locate where shape becomes visible.

A Proofs for the relation to operator induction

This appendix proves the results of Section 1 and states those that the main text mentions in one sentence. Results of our earlier report (Hänni and Claude (Anthropic), 2026a) and of the companion paper (Hänni and Claude (Anthropic), 2026b) are cited by their printed numbers and re-proved only where the setting changes. Throughout, $c, c', \ldots$ are machine constants. For a computable map h on indices, universality of V gives a string r_h with $V(r_h s) = h(V(s))$ for all $s \in \operatorname{dom} V$; so $p \mapsto r_h p$ is injective, maps $\operatorname{dom} V$ into $\operatorname{dom} V$ and adds $|r_h|$ to the length. "Tape by tape" means: for every tape ω.

A.1 Decompositions and the four maps

Measurability. $\{\omega : f_\omega \supseteq D\}$ is open. "f_ω is inconsistent" is open: an inconsistency uses finitely many labels, each certified under a prefix of ω, and these prefixes are comparable. "f_ω is closed"

is a countable intersection of Borel conditions. For theory machines $\{\omega : B \cup A_\omega \vdash \psi\}$ is open and consistency is closed (**??**). All events below are Borel.

Lemma A.1 (Mixture decompositions)**.** *For every finite $D \neq \emptyset$,*

$$M_F^{\mathrm{cons}}(D) = \sum_{p \in \operatorname{dom} V} 2^{-|p|} \nu^{\mathrm{cons}}_{F_{V(p)}}(D), \tag{1}$$

and the same holds for ν^{cl} (with M_F^{cl}) and for ν (with M_F). For $D = \emptyset$ add $1 - \Omega_V$ on the right for ν^{cons} and ν, and nothing for ν^{cl}. For every finite D, the empty one included,

$$M_A(D) = \sum_{p \in \operatorname{dom} V} 2^{-|p|} \nu_{G_{V(p)}}(D) + (1 - \Omega_V)\, \mathbf{1}[B \vdash \psi_D]. \tag{2}$$

Proof. $\operatorname{dom} V$ is prefix-free, so every tape has at most one prefix $p \in \operatorname{dom} V$. On a tape $p\omega'$, $f^{F_U}_{p\omega'} = f^{F_{V(p)}}_{\omega'}$ and $A^{G_U}_{p\omega'} = A^{G_{V(p)}}_{\omega'}$, with ω' uniform given p. On the tapes with no prefix in $\operatorname{dom} V$ (measure $1 - \Omega_V$), f_ω is nowhere defined, so it extends D only if $D = \emptyset$, and then it is consistent, B being consistent; it is never closed, because B proves every tautology and the empty labeller labels none. On these tapes $A_\omega = \emptyset$, so the theory is B, which is consistent and proves ψ_D iff $B \vdash \psi_D$. □

The weight of an index i is $\mathbf{m}(i) = \sum_{V(p)=i} 2^{-|p|}$, so (1) reads $M_F^{\mathrm{cons}} = \sum_i \mathbf{m}(i) \nu^{\mathrm{cons}}_{F_i}$ on nonempty D.

Lemma A.2 (The four maps)**.** *(i)* Proof search. *For a theory machine G let F_G answer a query φ as follows: in stages $s = 1, 2, \ldots$ read the tape up to bit s, run s steps of the enumeration of G restricted to pairs (w, χ) with w a prefix of the bits read, and test all derivations of size at most s from B and the sentences emitted so far; halt with 1 at the first derivation of φ and with 0 at the first derivation of $\neg\varphi$ (within a stage φ is tested first). Then F_G is a function machine, and on every tape ω with $B \cup A^G_\omega$ consistent, $f^{F_G}_\omega$ is consistent and closed and decides every sentence exactly as $B \cup A^G_\omega$ does.*

(ii) Emission. *For a function machine F let $G_F := \{(w, \varphi) : (\varphi, w, 1) \in F\} \cup \{(w, \neg\varphi) : (\varphi, w, 0) \in F\}$. Then $A^{G_F}_\omega = \Gamma_{f^F_\omega}$ for every ω. So if f^F_ω is consistent and extends D, then G_F is compatible with D on ω; if f^F_ω is closed and consistent, $B \cup A^{G_F}_\omega$ decides every sentence exactly as f^F_ω does.*

(iii) Craig's map with the tape. *For a function machine F, read as a machine that reads the tape one-way, let C_F answer a query χ as follows. Find the at most two decompositions $\chi = \psi^{\wedge m}$ (the right-nested m-fold conjunction: $m = 1$, or ψ the left conjunct of χ and m fixed by $|\chi|$). For each, test both readings: (1) run F on ψ for exactly $m - 1$ steps, reading the tape as F does, and accept if F halts at step $m-1$ with label 1; (2) if $\psi = \neg\varphi'$, run F on φ' for exactly $m - 1$ steps and accept if it halts at step $m - 1$ with label 0. Answer 1 if some test accepts and 0 otherwise. Then C_F is a random decider, and $\mathrm{Cn}(A^{\mathrm{dec}}_{C_F}(\omega)) = \mathrm{Cn}(\Gamma_{f^F_\omega})$ for every ω.*

(iv) Closure. *For a partial computable labeller f let $\bar{f}$ search for proofs of φ and of $\neg\varphi$ from $B \cup \Gamma_f$, dovetailing the enumeration of Γ_f. If f is consistent, $\bar{f}$ extends f, is consistent and closed, and $\mathrm{Cn}(\Gamma_{\bar{f}}) = \mathrm{Cn}(\Gamma_f)$.*

Proof. (i) The run on a query is a deterministic function of the bits read, and a halting run that has read w runs identically on every tape extending w; so the triples (φ, w, y) of halting runs form a consistent c.e. set. Let $B \cup A^G_\omega$ be consistent. If it proves φ^b, a derivation uses finitely many axioms, each emitted under a prefix of ω; at a late enough stage all of them have been enumerated and the derivation has size at most s, so the run halts, and it cannot first find a derivation of φ^{1-b}, since none exists. If the theory decides neither, no derivation is found and the label is undefined. So $f^{F_G}_\omega(\varphi) = b$ iff $B \cup A^G_\omega \vdash \varphi^b$. Then $\Gamma_f \subseteq \mathrm{Cn}(A^G_\omega)$ is consistent with B, and f is closed: $B \cup \Gamma_f \vdash \varphi^b$ implies $B \cup A^G_\omega \vdash \varphi^b$, which implies $f(\varphi) = b$; the converse holds as $\varphi^b \in \Gamma_f$. (ii) G_F is c.e. and emits φ^y exactly under the prefixes that certify $f(\varphi) = y$, so $A^{G_F}_\omega = \Gamma_{f^F_\omega}$. If f is closed, $B \cup \Gamma_f \vdash \varphi^b$ iff $f(\varphi) = b$. (iii) A decomposition $\chi = \psi^{\wedge m}$ with $m \geq 2$ has ψ the left conjunct and m fixed by $|\chi|$, since $|\psi^{\wedge m}| = m|\psi| + 3(m-1)$ in primitive syntax (Lemma E.12(b) of the earlier report); the only other candidate is $m = 1$. Every run is bounded by $m-1$ simulated steps, so C_F halts on every query and every tape: it is a random decider. Its answer depends only on the tape bits read by the simulated runs, so its triples are consistent. Its accepted set on ω is $\{\varphi^{\wedge(k+1)} : F \text{ halts on } \varphi \text{ with 1 in exactly } k \text{ steps}\} \cup \{(\neg\varphi)^{\wedge(k+1)} : F \text{ halts on } \varphi \text{ with 0 in exactly } k \text{ steps}\}$: reading (1) gives the first set (also when φ is itself a negation), reading (2) the second. Each member is equivalent to a member of $\Gamma_{f^F_\omega}$, and each member of $\Gamma_{f^F_\omega}$ has one, so Cn is preserved. (Reading (1) alone would lose the members $(\neg\varphi')^{\wedge(k+1)}$ that come from $f(\neg\varphi') = 1$, and Cn would change.) (iv) As (i), with the r.e. set Γ_f in place of A^G_ω; $f(\varphi) = b$ puts $\varphi^b \in \Gamma_f$, so $\bar{f}(\varphi) = b$. □

Only inclusions of events are claimed. The inclusion given by proof search can be strict: on a tape on which the theory is inconsistent, the labeller may already have committed to a consistent set of labels before the inconsistency appeared. A computation on a finite propositional analogue (code in the accompanying repository) finds many such tapes.

A.2 The equivalences

Proof of Theorem 1.2. We use coin-free forms of the maps of Lemma A.2: deterministic machines ignore the tape. *Enumerators to labellers* (proof search): $p \mapsto r_{\mathrm{ps}}p$; if the enumerator p is compatible with D, the proof-search labeller is consistent and closed, extends D, and decides each φ as the theory does. *Labellers to enumerators* (emission): $p \mapsto r_{\mathrm{em}}p$, the enumerator of $\Gamma_{\{V(p)\}}$; if the labeller is consistent and extends D, the theory is consistent and proves $D^{\pm}$; if it is closed, it decides as the labeller does. *Labellers to deciders* (Craig's map without tape): the decider of Craig's set of the labeller, which halts on every input even when the labeller is partial, since every run is bounded by the exponent of the candidate conjunction. *Deciders to enumerators*: enumerate the decided set; the theory is the same. *Consistent to closed*: the closure map. All maps are injective on programs and add a constant to the length. Summing $2^{-|p|}$ over compatible programs gives (a) and (c), where (c) uses proof search and emission restricted to closed labellers, which preserve the decision of every sentence; taking shortest programs gives (b). Finiteness in (b): if D is consistent, the lookup table of D is a consistent labeller, its closure a closed one, and the decider of the finite set $D^{\pm}$ is compatible; if not, no consistent theory proves $D^{\pm}$ and no consistent labeller extends D. For the losses, each of the two terms of $-\log W(D) + \log W(\emptyset)$ changes by at most c. For deciders this is Theorem 6.3 of the earlier report, stated there for programs that read their input; the change of program format costs a constant. □

With weights $2^{-\lambda|p|}$, $\lambda \geq 1$, as in the companion paper on steeper priors (Hänni and Claude (Anthropic), 2026c), the same maps give the equivalence within $2^{\pm\lambda c}$.

Remark A.3 (The schema route). A different route to Theorem 1.2 turns a labeller f into the axiom schema $A_f := \{\mathrm{Acc}_f(\ulcorner\varphi\urcorner) \to \varphi\} \cup \{\mathrm{Rej}_f(\ulcorner\varphi\urcorner) \to \neg\varphi\}$, with Σ_1 formulas Acc_f, Rej_f, written without $<$, that express that f accepts or rejects.

(a) *Two sorts.* Let $B = B_A \cup B_L$, with arithmetic axioms $\mathsf{Q} \subseteq B_A \subseteq \mathrm{Th}(\mathbb{N})$ on one sort, axioms B_L on the other and no mixed axioms. If $B_L \cup \Gamma_f$ is consistent, so is $B \cup A_f$, and the sentences of the non-arithmetic sort's language that $B \cup A_f$ proves are exactly those that $B_L \cup \Gamma_f$ proves (Proposition 6.4 of the earlier report; it uses the Σ_1-completeness of Q).

(b) *One sort over* PA. Assuming $\mathrm{Con}(\mathsf{PA})$, the labeller f that accepts $\neg\mathrm{Con}(\mathsf{PA})$ and, on $\mathrm{Con}(\mathsf{PA})$, searches for a PA-proof of $\bot$ has $\Gamma_f = \{\neg\mathrm{Con}(\mathsf{PA})\}$, which is consistent with PA, while $\mathsf{PA} \cup A_f$ is inconsistent (Proposition 6.5 there). So the argument through the schema fails in one sort; the conclusion of Theorem 1.2 does not.

(c) *Bounded witnesses.* In one sort, for every decidable $B \supseteq \mathsf{Q}$, true or not, let A_f^{bd} use the forms $\mathrm{Acc}_f^{\leq k}$, $\mathrm{Rej}_f^{\leq k}$ whose witnesses are bounded by a binary numeral k. Then $\mathrm{Cn}(A_f^{\mathrm{bd}}) = \mathrm{Cn}(\Gamma_f)$ for every labeller f, and membership in A_f^{bd} is decidable in polynomial time without running f (Proposition E.17 there, given the MRDP theorem): each axiom is Q-equivalent to a member of Γ_f or is a Q-theorem. In substance this is Craig's idea, the bound playing the role of the padding.

(d) *Templates.* No non-ground template of that report's template class has all its instances in the acceptance part of A_f; but for every Σ_n-sound labeller f one ground reflection sentence of length at most $a|f| + b_n$ reproduces f on Σ_n sentences over PA (Propositions 6.6 and 6.7 there, given partial truth predicates).

The one-tape equivalence avoids the schema altogether: a theory machine is an enumerator that reads a tape, so it may emit Γ_{f_ω} itself (emission); neither decidability, nor arithmetic, nor a schema is needed. The schema, like Craig's padding, matters only for the decider format.

Proof of Theorem 1.3. (b) is Lemma A.2(i)–(iii). (a), M_A against M_F^{cons}, for $D \neq \emptyset$: by (2), (b) and (1), with $p \mapsto r_{\mathrm{ps}}p$ (an index of $F_{G_{V(p)}}$) and $p \mapsto r_{\mathrm{em}}p$ (an index of $G_{F_{V(p)}}$),

$$\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq \sum_p 2^{-|p|}\nu^{\mathrm{cons}}_{F_{V(r_{\mathrm{ps}}p)}}(D) \leq 2^{|r_{\mathrm{ps}}|}M_F^{\mathrm{cons}}(D),$$
$$M_F^{\mathrm{cons}}(D) \leq 2^{|r_{\mathrm{em}}|}\sum_p 2^{-|p|}\nu_{G_{V(p)}}(D) \leq 2^{|r_{\mathrm{em}}|}M_A(D).$$

The remaining term $(1-\Omega_V)\mathbf{1}[B \vdash \psi_D]$ of (2) is at most $\mathbf{1}[B \vdash \psi_D]$; if $B \vdash \psi_D$, the deterministic proof-search labeller of B is closed and consistent and extends D on every tape, so $M_F^{\mathrm{cons}}(D) \geq 2^{-c_B}$. Hence $M_A \leq (2^{|r_{\mathrm{ps}}|} + 2^{c_B})M_F^{\mathrm{cons}}$. For $D = \emptyset$ both $M_A(\emptyset)$ and $M_F^{\mathrm{cons}}(\emptyset)$ lie in $[1-\Omega_V, 1]$ (the tapes without a program give the consistent theory B and the empty labeller), so their ratio is at most $1/(1-\Omega_V)$. M_A against M_A^{dec}: a random decider is a theory machine (emit χ under the prefix that certifies the answer 1), so $M_A^{\mathrm{dec}} \leq 2^c M_A$; and Craig's map $C_{F_{V(p)}}$ gives $M_A^{\mathrm{dec}} \geq 2^{-c}M_F^{\mathrm{cons}} \geq 2^{-2c}M_A$ on $D \neq \emptyset$; for $D = \emptyset$ the decider of $\emptyset$ gives $M_A^{\mathrm{dec}}(\emptyset) \geq 2^{-c}$. □

Corollary A.4 (Closed labellers and the trichotomy)**.** *For every finite D and $\varphi \notin \mathrm{dom}\, D$, the three outcome weights $M_A(D + (\varphi, 1))$, $M_A(D + (\varphi, 0))$, $M_A^3(D + (\varphi, \iota))$ (the theory proves φ, proves $\neg\varphi$, decides neither) agree within $2^{\pm c}$ with the measures of the tapes on which f_ω is closed, consistent, extends D and has $f_\omega(\varphi) = 1,\ 0,\ \bot$ respectively; and $2^{-c}M_A \leq M_F^{\mathrm{cl}} \leq M_F^{\mathrm{cons}}$.*

Proof. Proof search maps tapes with a consistent theory to tapes on which F_G is closed and decides as the theory does; emission maps tapes with a closed consistent labeller to tapes on which the theory decides as the labeller does. The program-less tapes (theory B) are covered by the closed deterministic labeller "proof search from B", of weight 2^{-c_B}, which decides as B does. Sum as in the proof of Theorem 1.3. The outcome weights of M_A are those of the corollary: $M_A(D + (\varphi, 1))$ is the measure of the tapes whose theory is consistent and proves ψ_D and φ, and likewise for the other two. □

Penalised versions. One may charge a hypothesis for the time it takes to check its axioms (a decider) or to generate them (an enumerator), with deduction still free (Definition A.7). The results on these versions rest on an assumption about the universal machine U whose programs are the hypotheses, taken from Appendix E.6 of the earlier report; it is standard for multitape machines, and we do not prove it, or its form with a tape, for a particular U.

Assumption A.5 ((S): efficient self-simulation)**.** Penalised hypotheses are programs of a fixed universal prefix machine U, read to their end before the input. There are constants $\alpha, a \geq 1$ such that the fixed wrappers used in the proofs read, copy and measure a program q within $\alpha(|q| + 1)$ steps; parse, compare or copy strings of total length L within $\alpha(L + 1)^2$ steps; and run the *dovetailing schedule* of q (at stage s, start q on the s-th sentence and advance every unfinished run by one simulated step) while keeping a *work counter* T (the simulated steps, plus the lengths of the sentences started, plus $|q|$), within $\alpha(T + 1)^a$ steps up to work T, writing a string of length L at cost at most αL. Put $a^* := \max(a, 3)$.

Assumption A.6 ((S$_{\text{tape}}$): self-simulation with a tape)**.** There are constants $\alpha, a \geq 1$ with the properties of Assumption A.5 when q also reads a one-way random tape: all runs of the dovetailing schedule read the same tape ω one-way from its start, the work counter also counts tape reads, and up to work T the schedule costs at most $\alpha(T + 1)^a$ steps and reads at most T tape bits.

We use the format of Appendix E.6 of the earlier report: hypotheses are programs of U, read to their end before the input, now with access to a one-way read-only random tape.

Definition A.7 (Penalised one-tape SAI)**.** A *random decider program* q halts on every pair (χ, ω) with yes or no; A^q_ω is its yes-set on ω, and $\mu^{\text{dec}}_q(D) := \mathbb{U}\{\omega : B \cup A^q_\omega$ consistent and $\vdash \psi_D\}$. A *random enumerator program* q has no input and writes sentences $a_1(\omega), a_2(\omega), \ldots$, completing a_i at step $\text{st}_i(\omega)$ ($\text{st}_0 := 0$). For time-constructible t:

- $M^{\text{chk},t}_A(D) := \sum_q 2^{-|q|}\mu^{\text{dec}}_q(D)$, the sum over the random decider programs q whose running time on every query χ and every tape is at most a constant multiple of $t(|\chi|)$, that is, $\sup_{\chi,\omega} \text{time}_q(\chi, \omega)/t(|\chi|) < \infty$;
- $M^{\text{gen},t}_A(D) := \sum_q 2^{-|q|}\nu^{\text{gen}}_q(D)$, with $\nu^{\text{gen}}_q(D) := \mathbb{U}\{\omega : B \cup \{a_i(\omega)\}_i$ consistent and $\vdash \psi_D\}$, the sum over the random enumerator programs that meet the *delay form* $\sup_{\omega,i}(\text{st}_i - \text{st}_{i-1})/t(|a_i|) < \infty$; in the *strict form* the condition is on $\text{st}_i/t(|a_i|)$, in the *cumulative form* on $\text{st}_i/t(|a_1| + \cdots + |a_i|)$.

The coin-free $W^{\text{chk},t}_A$, $W^{\text{gen},t}_A$ and the Levin-type $W^{\text{Kt},t}_A$ are those of Definition E.10 of the earlier report, which also assumes t nondecreasing and superadditive with $t(m) \geq m$.

Remark A.8 (On Assumption A.6). *Proof sketch.* We do not write this out for a concrete U; like Assumption A.5, the assumption is used, not proved. A tape-reading program can be simulated by keeping the bits read so far on a work tape and one pointer per run of the schedule; a simulated

read copies a stored bit or reads the next tape bit, at cost $O(T)$ up to work T. So the exponent a of Assumption A.5 should grow by at most 1.

In this format the unpenalised one-tape laws $M_A^U := \sum_q 2^{-|q|}\nu_{G_q}$, over all tape-reading programs read as theory machines, and the corresponding M_F^{cons} agree with M_A and M_F^{cons} within constant factors, since each universal mixture contains the other's universal machine behind a constant prefix.

Theorem A.9 (Axiom-side penalties change nothing)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible. There is c such that, for every finite D and in the strict, delay and cumulative forms: (a) under Assumption A.6, $2^{-c}M_F^{\mathrm{cons}}(D) \leq M_A^{\mathrm{chk},t}(D) \leq 2^c M_A(D)$ and $2^{-c}M_F^{\mathrm{cons}}(D) \leq M_A^{\mathrm{gen},t}(D) \leq 2^c M_A(D)$, the empty D included; so $M_A^{\mathrm{chk},t} \asymp M_A^{\mathrm{gen},t} \asymp M_A \asymp M_F^{\mathrm{cons}}$; (b) under Assumption A.5, $2^{-c}W^{\mathrm{cons}}(D) \leq W_A^{\mathrm{chk},t}(D) \leq W_A^{\mathrm{dec}}(D) \leq 2^c W^{\mathrm{cons}}(D)$ and $2^{-c}W^{\mathrm{cons}}(D) \leq W_A^{\mathrm{gen},t}(D) \leq 2^c W^{\mathrm{cons}}(D)$.*

Proof. (a) *Lower bounds.* Let p be a tape-reading program for a function machine F. Let q_F say yes on χ if F labels χ with 1, or if $\chi = \neg\varphi$ and F labels φ with 0 (dovetailing the two runs on the same tape). On every tape its yes-set is $\Gamma_{f_\omega^F}$, and $|q_F| \leq |p| + c$. The padded random decider d, a fixed wrapper followed by q_F, answers χ as in Lemma E.12(b) of the earlier report: it finds the at most two candidates $\chi = \psi^{\wedge m}$, rejects if $m-1 < |q_F|$, and otherwise runs the dovetailing schedule of q_F on the tape, without writing, until the work counter exceeds $m-1$, accepting iff q_F is first detected saying yes on ψ exactly at counter $m-1$. Under Assumption A.6 this takes at most $\alpha(|q_F|+1)+3\alpha(|\chi|+1)^{a'}$ steps, $a' := \max(a, 2)$, on every tape, uniformly in ω (the argument of that lemma, verbatim with the tape); as $(|\chi|+1)^{a'} \leq 2^{a'}|\chi|^{a^*} \leq 2^{a'}t(|\chi|)$, the membership bound holds. The answer depends only on the tape bits read by the schedule, so d is a random decider. On each tape its yes-set is $\mathrm{Pad}_\omega := \{\psi^{\wedge(T+1)} : q_F \text{ is first detected saying yes on } \psi \text{ at counter } T \text{ on } \omega\}$; distinct detections have distinct counters, so each $\psi \in \Gamma_{f_\omega^F}$ has exactly one padded copy and $\mathrm{Cn}(\mathrm{Pad}_\omega) = \mathrm{Cn}(\Gamma_{f_\omega^F})$ (Lemma E.12(a) there, per tape). So on every tape on which f_ω^F is consistent and extends D, d is compatible with D. The padded enumerator, a fixed wrapper followed by q_F, writes $\psi^{\wedge(T+1)}$ at each detection; Lemma E.12(c) there gives, on every tape, delay at most $2\alpha|a_i|^a$, strict completion at most $5\alpha|a_i|^{a^*}$ and cumulative completion at most $2\alpha(|a_1|+\cdots+|a_i|)^a$, uniformly in ω. Both maps are injective and add a constant to the length. Summing over p gives the lower bounds for $D \neq \emptyset$. For $D = \emptyset$, the decider that rejects every sentence runs in linear time with theory B, which is consistent, and the enumerator that writes nothing meets every delay condition vacuously; so $M_A^{\mathrm{chk},t}(\emptyset), M_A^{\mathrm{gen},t}(\emptyset) \geq 2^{-c} \geq 2^{-c}M_F^{\mathrm{cons}}(\emptyset)$. *Upper bounds.* A random decider program q is read as the theory machine that emits χ under the tape prefix read by a run of q on χ that answers yes; a random enumerator program as the theory machine that emits the i-th written sentence under the prefix read when it is completed. Each reading is a fixed computable map on programs, realised by a constant-size wrapper, which is injective and keeps the theory on every tape. So $M_A^{\mathrm{chk},t}, M_A^{\mathrm{gen},t} \leq 2^c M_A^U \leq 2^{c'} M_A$ for every D. With Theorem 1.3 this gives the chain of equivalences. (b) This is Theorem 6.14(a) of the earlier report: the coin-free version of the construction above, with the padded sets of Lemma E.12 there. The inequality $W_A^{\mathrm{chk},t} \leq W_A^{\mathrm{dec}}$ holds because the penalised deciders are deciders with the same weights, and $W_A^{\mathrm{dec}} \leq 2^c W^{\mathrm{cons}}$ by Theorem 1.2. □

The equivalence also holds class by class (coin-free, Proposition A.28).

Proposition A.10 (Class posteriors, one tape)**.** *Let $t(m) \geq m^{a^*}$ be time-constructible and X be $M_A^{\mathrm{chk},t}$ or $M_A^{\mathrm{gen},t}$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C}; D)$ be the*

measure, under X's mixture, of the tapes on which the hypothesis is compatible with D and the deductive closure of its axioms is in $\mathcal{C}$. Under Assumption A.6 there is c with $2^{-c}\mathbb{U}_A(\mathcal{C};D) \le X(\mathcal{C};D) \le 2^c\mathbb{U}_A(\mathcal{C};D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of theories (genuine axiom systems and disguised labellers alike) under X and under M_A differ by a factor at most 2^{2c}.

Proof. Lower bound: for a theory machine G with program p, let q_G say yes on χ iff G emits χ on the tape, and pad as in the proof of Theorem A.9. On every tape the padded hypothesis has the same deductive closure as $B \cup A^G_\omega$, so it lies in $\mathcal{C}$ and is compatible with D exactly when G's theory is. The program-less tapes of M_A (theory B) are matched by the decider of $\emptyset$. Upper bound: a penalised hypothesis, read as a theory machine, has the same theory. For the odds, divide by $X(\text{all};D)$ and $\mathbb{U}_A(\text{all};D) = M_A(D)$. □

Proposition A.11 (The chain for the consistent versions)**.** *For every consistent $D \neq \emptyset$,*

$$2^{-K^{\text{cons}}_F(D)} \le W^{\text{cons}}(D) \le 2^c M^{\text{cons}}_F(D) \le 2^{K(\operatorname{dom} D)+c}\, 2^{-K^{\text{cons}}_F(D)},$$

and $\log(M^{\text{cons}}_F(D)/W^{\text{cons}}(D)) \le K^{\text{pad}}(L^{\text{cons}}_{1/2}(D)) + c$, where $L^{\text{cons}}_{1/2}(D)$ is the least L with $\mathbb{U}\{\omega :$ every label of D is certified within $\omega_{1:L}$ and f_ω is consistent$\} \ge M^{\text{cons}}_F(D)/2$. By Theorems 1.2 and 1.3 the same holds for K_A, W_A, M_A with a larger c.

Proof. A shortest consistent program extending D is a term of W^{cons}. A consistent deterministic labeller, read as a function machine that ignores the tape, is consistent on every tape; with (1) and the wrapper "treat $\{V(p)\}$ as a machine", $W^{\text{cons}} \le 2^c M^{\text{cons}}_F$. Next, $M^{\text{cons}}_F \le M_F \le 2^{-K(D|\operatorname{dom} D)+c}$ by Theorem 3.21(a) of the companion paper, which holds for the input set Sent; the lookup table of D is consistent, so $K^{\text{cons}}_F(D) \le K(D) + c \le K(\operatorname{dom} D) + K(D \mid \operatorname{dom} D) + c$. For the K^{pad} bound, take $L' \ge L$ with $K(L') = K^{\text{pad}}(L)$, and for $w \in \{0,1\}^{L'}$ the stored-prefix labeller $\varphi \mapsto y$ iff some $(\varphi, w', y) \in F$ has $w' \sqsubseteq w$. It is a sub-function of f^F_ω for every $\omega \sqsupseteq w$, so it is consistent whenever some extension of w is a consistent tape. The count in the proof of Proposition 4.6(a) of the companion paper, restricted to such w, gives, with r the constant-size wrapper, $W^{\text{cons}}(D) \ge 2^{-|r|-K(F)-K^{\text{pad}}(L)}\mathbb{U}\{\omega : D$ certified within $\omega_{1:L}$, f^F_ω consistent$\}$; take $F = F_U$ and $L = L^{\text{cons}}_{1/2}(D)$. □

A.3 Positive literals

Proof of Theorem 1.4. By Theorems 1.2 and 1.3 it suffices to compare consistent SOI on sentences with plain SOI on $\mathcal{I}$. *Strings to sentences:* for a partial computable g on $\mathcal{I}$ let $g^P(R(t_x)) := g(x)$, undefined off the literals, and likewise for a function machine on $\mathcal{I}$ (decode x from t_x). Γ_{g^P} is a set of literals $\pm R(t_x)$, at most one per x, and it is consistent with B: in a model of B in which the t_x are pairwise distinct, the symbol R, which B does not mention, can be interpreted as $\{t_x : g(x) = 1\}$. So g^P, and the machine version on every tape, is consistent, extends D iff g extends D^{str}, and costs a constant. *Sentences to strings:* $f^{\text{str}}(x) := f(R(t_x))$, at a constant cost; $f \supseteq D$ implies $f^{\text{str}} \supseteq D^{\text{str}}$, and consistency is not needed. Sum over the injective maps, and take shortest programs. For the three outcome weights of $R(t_y)$: a consistent set of literals of this kind proves exactly its members (for $R(t_y)^b$ not in the set, reinterpret R at t_y), so the theory $\text{Cn}(\Gamma_{g^P})$ decides $R(t_y)$ iff $g(y)$ is defined, with the same value; and the proof-search labeller of a theory, restricted to literals, labels y exactly as the theory decides $R(t_y)$. □

Corollary A.12 (What transfers)**.** *In the setting of Theorem 1.4, every result of the companion paper about binary labels that holds up to additive constants in K, $-\log W$ and $-\log M_F$ holds for SAI on literal data, with the same quantities. In particular:*

(a) no coding theorem for axiom systems*: for every $d \geq 2$ there are distinct $x_1, x_2 \in \{0,1\}^d$ such that $D := \{(R(t_{x_1}), 0), (R(t_{x_2}), 1)\}$ has $K_A(D) \geq \log d - 2 - c$, $-\log W_A(D) \leq 2 + K(\lceil \log(d+1) \rceil) + c$ and $-\log M_A(D) \leq 2 + c$; and over two-point literal data $\log d - c \leq \max_D (K_A(D) - L_A(D)) \leq \log d + O(\log\log d)$ (Theorem 4.9 and Corollary 4.10 of the companion paper); **??** proves the first three bounds directly;*

(b) random theories beat deterministic ones*: $\sup_D M_A(D)/W_A(D) = \infty$ over two-point literal data, and $W_A(D) \geq 2^{-|D| - K^{\mathrm{pad}}(2^{d+1}) - c}$ for every literal data set over words of length at most d (Propositions 4.5(a) and 4.6 there);*

(c) shared randomness is not a mixture of private coins*, read as a comparison of M_A on literal data D with the companion paper's product mixture M_{prod} on D^{str}: M_A/M_{prod} is unbounded (Proposition 4.7 there); no product law internal to SAI is defined here;*

(d) Solomonoff induction*: with unary numerals $t_i := S^i 0$ and $B \in \{\emptyset, \mathsf{Q}\}$, $M_A(\{(R(S^i 0), x_{i+1}) : i < n\})$ is within $2^{\pm c}$ of the universal semimeasure of $x \in \{0,1\}^n$ (Proposition 3.24 there).*

Proof. Theorem 1.4 changes K, $-\log W$ and $-\log M_F$ by at most a constant, so every statement that holds up to additive constants in these quantities transfers. (a) is Theorem 4.9 and Corollary 4.10 of the companion paper applied to D^{str}; (b) is its Propositions 4.5(a) and 4.6; (c) is its Proposition 4.7, read through $M_A(D) \asymp M_F(D^{\mathrm{str}})$, so that it compares M_A with M_{prod} on D^{str}; (d) is its Proposition 3.24 with $t_i := S^i 0$, which are pairwise distinct in $\mathbb{N}$, a model of $B \in \{\emptyset, \mathsf{Q}\}$ (with R interpreted arbitrarily). The statements that do not transfer are of the following kinds: labels from a set larger than $\{0,1\}$ (a fresh unary predicate carries one bit); cumulative statements in the normalised convention (each normalised conditional transfers only within $2^{\pm 2c}$, so n steps cost $2cn$; single-step quantities do transfer up to a constant factor); private randomness per input or per query, and repeated inputs with fresh noise (no mixture over theory machines with a private coin per atom is defined here, and conflicting labels on one sentence are inconsistent data); history-aware and weak-semimeasure variants, which have no analogue here; computability statements, which are not "up to constants"; and other objects (time-bounded mixtures, circuits), which we compare directly (Theorems A.29 and A.43). □

Proposition A.13 (The price of the coin on literal data)**.** *In the setting of Theorem 1.4, for every literal data set D over words of length at most d,*

$$\log \frac{M_A(D)}{W_A(D)} \leq \delta_D + K^{\mathrm{pad}}(2^{d+1}) + c \leq \delta_D + K(\lceil \log(d+1) \rceil) + c', \qquad \delta_D := |D| - K(D^{\mathrm{str}} \mid \operatorname{dom} D^{\mathrm{str}}).$$

Proof. $W_A(D) \geq 2^{-c} W(D^{\mathrm{str}}) \geq 2^{-|D| - K^{\mathrm{pad}}(2^{d+1}) - c'}$ by Theorem 1.4 and Proposition 4.6(b) of the companion paper, and $M_A(D) \leq 2^c M_F(D^{\mathrm{str}}) \leq 2^{-K(D^{\mathrm{str}} | \operatorname{dom} D^{\mathrm{str}}) + c''}$ by its Theorem 3.21(a). □

So on literal data whose labels are algorithmically random given the inputs ($\delta_D = O(1)$), the coin-free version loses at most $K^{\mathrm{pad}}(2^{d+1}) + O(1)$ bits against the one-tape version, however many labels are predicted.

Remark A.14 (An encoding effect). Let χ_i be $R(t_{w_i})$ or $\neg R(t_{w_i})$ by independent fair coins, for distinct words w_i, with data $(\chi_i, 1)$, $i \leq n$. Plain coin-free SOI on sentences, W^{all}, has bounded loss, through the labeller "always 1", while SAI and consistent SOI, coin-free or one-tape, lose at

least n bits in expectation (Proposition E.1 of the earlier report). This is an encoding effect: the map $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ turns these data into fair-coin labels on positive literals and leaves M_A and W_A unchanged, since they depend on D only through $\mathrm{Cn}(D^{\pm})$ (**??**); there SAI is SOI on strings, which loses at least $n - c$ bits in expectation on fair coins. The labeller "always 1" is inconsistent; consistency is what ties $\neg R(t_w)$ to $R(t_w)$.

Proof. The lower bound for SAI and consistent SOI is Proposition E.1 of the earlier report: on every tape the theory (or the consistent labeller) cannot accept both $R(t_w)$ and $\neg R(t_w)$, so the two predictive probabilities sum to at most 1, and convexity of $-\log$ gives at least one bit per step in expectation; the argument applies tape by tape to M_A and M_F^{cons}. The re-encoding $(\neg R(t_w), 1) \mapsto (R(t_w), 0)$ is a bijection of data sets that keeps $D^{\pm}$, so it keeps M_A and W_A (**??**). On positive literals SAI is SOI on strings (Theorem 1.4), and on fair-coin labels the latter loses at least $n - c$ bits in expectation, by Theorem 3.21(a) of the companion paper and Gibbs' inequality. □

A.4 What breaks

Proof of Proposition 1.5. (a) A consistent theory proving ψ_D proves χ; and M_A depends on D only through $\mathrm{Cn}(D^{\pm})$ (**??**). (b) A labeller $f \supseteq D$ with $f(\chi) = 0$ has $D^{\pm} \cup \{\neg\chi\} \subseteq \Gamma_f$, which is inconsistent with B. By Theorem 1.2, $W^{\mathrm{cons}}(D + (\chi, 1)) \geq 2^{-c} W_A(D + (\chi, 1)) = 2^{-c} W_A(D) \geq 2^{-2c} W^{\mathrm{cons}}(D)$; for one tape use Theorem 1.3. (c) Let q_χ be a shortest V-program for χ. The prefix machine that reads q_χ, then $s \in \operatorname{dom} V$, and outputs an index of "$\{V(s)\}$, made undefined at $V(q_\chi)$" is simulated by V with a prefix r. The map $s \mapsto r q_\chi s$ is injective; removing one label preserves consistency, and the restricted labeller still extends D because $\chi \notin \operatorname{dom} D$. So the consistent labellers extending D and undefined at χ have weight at least $2^{-|r|-K(\chi)} W^{\mathrm{cons}}(D)$. For one tape, apply the same map to function machines and use (1); on the program-less tapes the empty labeller is already undefined at χ. (d) is Theorem 1.2(c) and Corollary A.4. □

Remark A.15 (The three outcomes are a measure for SAI). For $\varphi \notin \operatorname{dom} D$, $M_A(D + (\varphi, 1)) + M_A(D + (\varphi, 0)) + M_A^3(D + (\varphi, \iota)) = M_A(D)$ (**??**), and likewise for W_A: each consistent theory proving ψ_D falls in exactly one of the three classes. For SOI the corresponding third mass is the defect, which collects abstention and divergence alike.

Productivity of the independent sentences. Ref_B and Thm_B are the sets of B-refutable and B-provable sentences. We use three textbook facts about Q (standard; see Boolos et al., 2007): (R) every decidable relation on numbers is strongly represented in Q, by a formula $\theta(\bar{x})$ with $\mathsf{Q} \vdash \theta(\underline{\bar{n}})$ when the relation holds of $\bar{n}$ and $\mathsf{Q} \vdash \neg\theta(\underline{\bar{n}})$ when it does not; (O) for every n, $\mathsf{Q} \vdash \forall z\,(z \leq \underline{n} \to z = \underline{0} \vee \cdots \vee z = \underline{n})$ and $\mathsf{Q} \vdash \forall y\,(y \leq \underline{n} \vee \underline{n} \leq y)$, with $\leq$ the usual abbreviation; (Diag) for every formula $\vartheta(x)$ there is a sentence ρ, computable from ϑ, with $\mathsf{Q} \vdash \rho \leftrightarrow \vartheta(\ulcorner \rho \urcorner)$.

Lemma A.16 (Productivity of the independent sentences)**.** *Let $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$, and let B be decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$. From an index of an r.e. set $S \subseteq \mathrm{Ind}_B$ one computes an $\mathcal{L}_{\mathrm{ar}}$-sentence $\rho \in \mathrm{Ind}_B \setminus S$. In particular Ind_B is productive, hence not r.e., and Ref_B is r.e. and not decidable.*

The lemma is classical: it is a form of the effective inseparability of the provable and the refutable sentences of consistent r.e. extensions of Q (cf. Smullyan, 1961). We include the proof, which is Rosser's argument relative to S.

Proof. Let $\mathsf{Neg}(y,x)$ be the decidable relation "y codes a B-derivation of the negation of the sentence coded by x, or x is enumerated into S within y steps", and $\mathsf{Prf}(y,x)$ the decidable relation "y codes a B-derivation of the sentence coded by x" (B is decidable). Let $\theta_{\mathrm{neg}}, \theta_{\mathrm{prf}}$ strongly represent them (R), and let ρ satisfy $\mathsf{Q} \vdash \rho \leftrightarrow \exists y\,(\theta_{\mathrm{neg}}(y, \ulcorner\rho\urcorner) \wedge \forall z \le y\ \neg\theta_{\mathrm{prf}}(z, \ulcorner\rho\urcorner))$ (Diag); ρ is computed uniformly from the index of S. $B \nvdash \rho$. Suppose k codes a B-derivation of ρ. Then $\rho \notin \mathrm{Ind}_B \supseteq S$, and $B \nvdash \neg\rho$ by consistency, so $\mathsf{Neg}(n, \ulcorner\rho\urcorner)$ fails for every n and $\mathsf{Q} \vdash \neg\theta_{\mathrm{neg}}(\underline{n}, \ulcorner\rho\urcorner)$ for every n (R); also $\mathsf{Q} \vdash \theta_{\mathrm{prf}}(\underline{k}, \ulcorner\rho\urcorner)$. Argue in Q: given y with $\theta_{\mathrm{neg}}(y, \ulcorner\rho\urcorner) \wedge \forall z \le y\, \neg\theta_{\mathrm{prf}}(z, \ulcorner\rho\urcorner)$, by (O) either $\underline{k} \le y$, which contradicts $\theta_{\mathrm{prf}}(\underline{k}, \ulcorner\rho\urcorner)$, or $y \le \underline{k}$, so that y is one of $\underline{0}, \ldots, \underline{k}$, which contradicts $\neg\theta_{\mathrm{neg}}(\underline{n}, \ulcorner\rho\urcorner)$. So $\mathsf{Q} \vdash \neg\rho$, hence $B \vdash \neg\rho$, and B is inconsistent, a contradiction. $B \nvdash \neg\rho$ *and* $\rho \notin S$. Otherwise $\mathsf{Neg}(n, \ulcorner\rho\urcorner)$ holds for some n, while $\mathsf{Prf}(m, \ulcorner\rho\urcorner)$ fails for every m by the previous step. By (R) and (O), $\mathsf{Q} \vdash \theta_{\mathrm{neg}}(\underline{n}, \ulcorner\rho\urcorner) \wedge \forall z \le \underline{n}\, \neg\theta_{\mathrm{prf}}(z, \ulcorner\rho\urcorner)$, so $\mathsf{Q} \vdash \rho$ and $B \vdash \rho$, against the previous step. So $\rho \in \mathrm{Ind}_B \setminus S$. A productive set is not r.e. (take S to be the set itself). Ref_B is r.e. since B is decidable; if it were decidable, so would be $\mathrm{Thm}_B = \{\varphi : \neg\varphi \in \mathrm{Ref}_B\}$, hence Ind_B, which is not even r.e. □

Proof of Proposition 1.6. (a) Coin-free: map a consistent three-valued labeller f to the enumerator of Γ_f (its 1/0 part); the theory $\mathrm{Cn}(\Gamma_f)$ proves the 1/0 data and, by the consistency of f, decides none of the sentences f labels ι, which include those of D. One tape: emission, tape by tape, on the tapes on which f^F_ω is consistent and extends D, and (1). (b) Coin-free: let $\mathcal{F}_n$ be the finite (not computable) set of the $p \in \mathrm{dom}\, V$ with $|p| \le n$ for which $\{V(p)\}$ is a consistent three-valued labeller, and I_n the union of their ι-sets. It is a finite union of r.e. sets, hence r.e., and $I_n \subseteq \mathrm{Ind}_B$, since $f(\sigma) = \iota$ for a consistent f implies $B \cup \Gamma_f \nvdash \sigma, \neg\sigma$. Let $\sigma_n \in \mathrm{Ind}_B \setminus I_n$ (Lemma A.16). Every consistent labeller extending $\{(\sigma_n, \iota)\}$ labels σ_n with ι, so its program is not in $\mathcal{F}_n$ and is longer than n; hence $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \le \Omega_V^{\ge n+1}$. The enumerator of $\emptyset$ (theory B) is compatible with $\{(\sigma_n,\iota)\}$, so $W^3_A(\{(\sigma_n,\iota)\}) \ge 2^{-c}$. One tape: fix n and $\varepsilon := 2^{-n}$. For $p \in \mathrm{dom}\, V$ with $|p| \le n$ let $F_p := F_{V(p)}$ and Cons_p the (Borel) set of tapes on which $f^{F_p}_\omega$ is consistent. If $\mathbb{U}(\mathrm{Cons}_p) = 0$, p contributes nothing. Otherwise, by the Lebesgue density theorem (**??**) almost every $\omega \in \mathrm{Cons}_p$ has a prefix w with $\mathbb{U}(\mathrm{Cons}_p \cap [w]) > (1-\varepsilon)2^{-|w|}$. Let $\mathcal{W}_p$ be the set of minimal such prefixes, a prefix-free set with $\mathbb{U}(\mathrm{Cons}_p \setminus \bigcup_{w\in\mathcal{W}_p}[w]) = 0$, and choose a finite $\mathcal{W}'_p \subseteq \mathcal{W}_p$ with $\sum_{w\in\mathcal{W}_p\setminus\mathcal{W}'_p} 2^{-|w|} \le \varepsilon$. For $w \in \mathcal{W}'_p$ put

$$J_w := \{\sigma : \mathbb{U}([w] \cap \{\omega : f^{F_p}_\omega(\sigma) = \iota\}) > \varepsilon 2^{-|w|}\}.$$

J_w is r.e., uniformly in (p,w) and the rational ε, because the measure of the open set $[w]\cap\{f(\sigma) = \iota\}$ is lower semicomputable. And $J_w \subseteq \mathrm{Ind}_B$: if $\sigma \in J_w$, the set $[w] \cap \{f(\sigma) = \iota\}$ has measure larger than $\varepsilon 2^{-|w|} > \mathbb{U}([w] \setminus \mathrm{Cons}_p)$, so it meets Cons_p; on such a tape f_ω is consistent and labels σ with ι, so $B \cup \Gamma_{f_\omega}$, and hence B, decides neither σ nor $\neg\sigma$. Let $I'_n := \bigcup_{|p|\le n}\bigcup_{w\in\mathcal{W}'_p} J_w$, a finite union of r.e. subsets of Ind_B, and $\sigma'_n \in \mathrm{Ind}_B \setminus I'_n$ (Lemma A.16). For $|p| \le n$, since $\sigma'_n \notin J_w$,

$$\begin{aligned}\nu^{3,\mathrm{cons}}_{F_p}(\{(\sigma'_n,\iota)\}) &\le \sum_{w\in\mathcal{W}'_p} \mathbb{U}([w] \cap \{f(\sigma'_n) = \iota\}) + \sum_{w\in\mathcal{W}_p\setminus\mathcal{W}'_p} 2^{-|w|} + \mathbb{U}\Big(\mathrm{Cons}_p \setminus \bigcup_{w\in\mathcal{W}_p}[w]\Big)\\ &\le \sum_{w\in\mathcal{W}'_p} \varepsilon 2^{-|w|} + \varepsilon \le 2\varepsilon.\end{aligned}$$

By (1), $M^{3,\mathrm{cons}}_F(\{(\sigma'_n,\iota)\}) \le \sum_{|p|\le n} 2^{-|p|}2\varepsilon + \Omega_V^{\ge n+1} \le 2^{1-n} + \Omega_V^{\ge n+1}$. The deterministic theory machine that emits nothing (theory B) is compatible with $\{(\sigma'_n,\iota)\}$ on every tape, so $M^3_A(\{(\sigma'_n,\iota)\}) \ge 2^{-c}$. (c) For $\sigma \in \mathrm{Ind}_B$ the labeller "ι at σ, undefined elsewhere" is consistent and has length

$K(\sigma)+c$; with $W^3_A \le 1$ this gives the coin-free bound, and as a function machine that ignores the tape it gives the one-tape bound. □

The coin-free part of (b) is the special case of machines that ignore the tape: then J_w is the labeller's ι-set and no density argument is needed. One tape, the obstacle is that the set of sentences a random machine labels ι on consistent tapes need not be r.e., since it depends on the positivity of the measure of a Π^0_1 class; the density point restores an r.e. subset of Ind_B that carries almost all of the consistent ι-mass. A computation on finite analogues (code in the accompanying repository), with machines consistent on most tapes and noise that labels B-decided sentences ι, confirms the density step and the summed bound in every case tested.

Proposition A.17 (Large gaps have no r.e. witnesses)**.** *In the setting of Proposition 1.6: (a) for every r.e. $S \subseteq \mathrm{Ind}_B$ and every $\sigma \in S$, $W^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \ge 2^{-K(S)-c}$ and $M^{3,\mathrm{cons}}_F(\{(\sigma,\iota)\}) \ge 2^{-K(S)-c}$, with $K(S)$ the complexity of an index of S; so no computable sequence $\sigma_n \in \mathrm{Ind}_B$ has $W^{3,\mathrm{cons}}(\{(\sigma_n,\iota)\}) \to 0$. (b) If $n(\sigma)$ is the length of a shortest program of a consistent three-valued labeller that labels σ with ι, then $2^{-n(\sigma)} \le W^{3,\mathrm{cons}}(\{(\sigma,\iota)\}) \le \Omega_V^{\ge n(\sigma)}$.*

Proof. (a) The labeller "ι on every member of S (enumerate S), undefined elsewhere" has $\Gamma = \emptyset$ and labels only B-independent sentences, so it is consistent; it has length $K(S)+c$. As a function machine that ignores the tape it gives the one-tape bound. (b) Every program counted has length at least $n(\sigma)$, and one has length $n(\sigma)$. □

Remark A.18 (Shared versus private randomness). If a product law (private coins per input, Proposition 3.9(a) of the companion paper) gives positive probability to labelling φ true and to labelling ψ true, with $B \cup \{\varphi,\psi\}$ inconsistent, the sampled labeller is inconsistent with probability at least the product of the two. The consistent part of a product law is not a product law: for a B-independent φ, on the two inputs φ, $\neg\varphi$ with fair independent labels, the consistent outcomes are $(1,0)$ and $(0,1)$, each of mass $\frac{1}{4}$, while a product would need $P(1,1)P(0,0) = P(1,0)P(0,1) \ne 0$. (If B decides φ, one of these two outcomes is inconsistent too.) The shared tape is what makes a random labeller a random *theory*; private randomness is harmless on logically independent inputs such as positive literals (Theorem 1.4). Product laws are machine laws, so M^{cons}_F dominates their consistent parts.

Proposition A.19 (Computability of the versions)**.** *(a) Suppose the set of finite data sets D with $B \cup D^{\pm}$ consistent is not r.e. (for instance if $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ and B is decidable and consistent with $\mathsf{Q} \subseteq \mathrm{Cn}(B)$, since then $\{\varphi : \{(\varphi,1)\}$ is consistent$\}$ is the complement of Ref_B, which is r.e. and not decidable by Lemma A.16; or if $B = \emptyset$ and $\mathcal{L}$ has a binary relation symbol, by Church's theorem, Church, 1936). Then none of W_A, W^{dec}_A, W^{cons}, W^{cl}, M_A, M^{dec}_A, M^{cons}_F, M^{cl}_F is lower semicomputable.*

(b) In every setting, $M_A(\emptyset)$ and $M^{\mathrm{cons}}_F(\emptyset)$ are not left-c.e.; so M_A and M^{cons}_F are not lower semicomputable.

(c) W_A, W^{cons}, M_A and M^{cons}_F are computable relative to $\emptyset'$, uniformly in D.

(c′) W^{dec}_A, M^{dec}_A, W^{cl} and M^{cl}_F are computable relative to $\emptyset''$, uniformly in D.

Whether the four versions of (c′) are computable relative to $\emptyset'$ is not settled here. Being closed and being a (random) decider are Π_2 conditions, and for closedness the condition is Π_2-complete: with $\chi_0, \chi_1, \ldots$ an injective enumeration of the B-decided sentences, the labeller that labels χ_n with B's decision once program e halts on n is consistent, and it is closed iff e is total.

Proof. (a) For a function ν that is lower semicomputable uniformly in D, the support $\{D : \nu(D) > 0\}$ is r.e. Each object listed has support exactly the consistent data sets: positive weight needs a consistent hypothesis that proves or extends D; conversely, for a consistent D the lookup table of D (as a labeller, a finite theory, a decider, or a machine that ignores the tape) has positive weight in W_A, W_A^{dec}, W^{cons}, M_A, M_A^{dec} and M_F^{cons}, and for the closed versions W^{cl}, M_F^{cl} its closure "proof search from $B \cup D^{\pm}$", which is closed, consistent and extends D, has positive weight (the lookup table itself is never closed). (b) For M_A this is **??**. For M_F^{cons}: for a left-c.e. real ζ with approximations $\zeta_s \uparrow \zeta$, let F_ζ label the sentence $\exists x\, x \neq x$ with 1 under a prefix w as soon as some stage s has $0.w + 2^{-|w|} \leq \zeta_s$, and be undefined elsewhere; its labelling is inconsistent with probability $\mathbb{U}\{\omega : 0.\omega < \zeta\} = \zeta$. With p_ζ a V-program for F_ζ, the probability that f_ω is inconsistent is $2^{-|p_\zeta|}\zeta + \kappa$ with κ left-c.e. If $M_F^{\mathrm{cons}}(\emptyset)$, which is one minus that probability, were left-c.e., the probability would be computable and ζ right-c.e.; take $\zeta = \Omega_V$. (c) W_A and W^{cons} are sums $\sum_p 2^{-|p|}\mathbf{1}[\cdot]$ of Δ_2 indicators ("proves ψ_D" and "extends D" are Σ_1, consistency is Π_1), and the tails $\Omega_V^{\geq \ell}$ are $\emptyset'$-computable; so $\emptyset'$ computes each value to any precision. For one tape, $M_A(D) = \mathbb{U}(O_{\psi_D}) - \mathbb{U}(O_\perp)$ (**??**), and $M_F^{\mathrm{cons}}(D) = \mathbb{U}(O) - \mathbb{U}(O \setminus C)$ with $O := \{\omega : f_\omega \supseteq D\}$ effectively open and $C := \{\omega : f_\omega \text{ consistent}\}$ effectively closed: differences of left-c.e. reals, uniformly in D. (c′) Coin-free: "$V(p)$ is a total $\{0,1\}$-valued program" and "$\{V(p)\}$ is closed" $(\forall \varphi, b : B \cup \Gamma \vdash \varphi^b \to \{V(p)\}(\varphi) = b)$ are Π_2, so $\emptyset''$ decides each indicator and computes the sums. One tape, deciders: "$F_{V(p)}$ halts on every query and every tape" is Π_2, because for each query, halting on all tapes is, by compactness of Cantor space, equivalent to the Σ_1 condition that some n and s make every tape prefix of length n halt within s steps. For a random decider $\mu_F^{\mathrm{dec}}(D)$ is a difference of left-c.e. reals uniformly in an index, as in (c); so $\emptyset''$ computes $M_A^{\mathrm{dec}}(D)$. One tape, closed: "f_ω is closed" is the countable intersection over (φ, b) of the sets $\{\omega : B \cup \Gamma_{f_\omega} \nvdash \varphi^b\} \cup \{\omega : f_\omega(\varphi) = b\}$, each the union of an effectively closed and an effectively open set, hence an effective G_δ. So the event in $M_F^{\mathrm{cl}}(D)$ is an effective Π_2^0 class $\bigcap_k U_k$ with U_k uniformly effectively open and decreasing; its measure $\inf_k \mathbb{U}(U_k)$ is an infimum of uniformly left-c.e., hence uniformly $\emptyset'$-computable, reals, so it is right-c.e. relative to $\emptyset'$ and computable from $\emptyset''$. □

Question A.20. Can W_A be lower semicomputable when consistency of finite sets of sentences with B is decidable (for instance for a propositional language, or a monadic language without function symbols and $B = \emptyset$)?

There the support argument of Proposition A.19(a) says nothing, and the argument of (b) does not carry over directly to the coin-free version, because the weight of the inconsistent programs is not separated from Ω_V by a computable amount.

Pathwise and machine-wise consistency. M_F^{cons} conditions on the event that the *sampled* labeller is consistent. The machine-wise alternative is

$$M_F^{\mathrm{mw}}(D) := \sum_{p \in \mathrm{dom}\, V} 2^{-|p|}\, \mathbf{1}[F_{V(p)} \text{ is consistent on every tape}]\, \nu_{F_{V(p)}}(D),$$

and M_A^{mw} is defined likewise over the theory machines whose theory is consistent on every tape.

Proposition A.21 (Pathwise against machine-wise consistency)**.** *(a)* $W^{\mathrm{cons}} \leq 2^c M_F^{\mathrm{mw}} \leq 2^c M_F^{\mathrm{cons}}$ *on nonempty* D*, and* $M_F^{\mathrm{mw}} \asymp M_A^{\mathrm{mw}}$.

(b) For every consistent $D \neq \emptyset$, $M_F^{\mathrm{cons}}(D) \leq 2^{\min\{K^{\mathrm{pad}}(L_{1/2}^{\mathrm{cons}}(D)),\, K(\mathrm{dom}\, D)\}+c}\, M_F^{\mathrm{mw}}(D)$*, with* $L_{1/2}^{\mathrm{cons}}$ *as in Proposition A.11.*

(c) *If consistency of finite sets with B is decidable (an algorithm decides, for finite $\Lambda \subseteq \mathrm{Sent}$, whether $B \cup \Lambda$ is consistent; for example if $\mathrm{Cn}(B)$ is decidable, for propositional languages, and for monadic languages without function symbols with $B = \emptyset$, by the decidability of monadic first-order logic, Löwenheim, 1915; Behmann, 1922), then $M_F^{\mathrm{cons}} \asymp M_F^{\mathrm{mw}}$ and $M_A \asymp M_A^{\mathrm{mw}}$.*

Proof. (a) A deterministic consistent labeller is consistent on every tape, and a machine consistent on every tape has $\nu = \nu^{\mathrm{cons}}$. Proof search and emission preserve "consistent on every tape", since on each tape proof search of a consistent theory is consistent and emission of a consistent labelling is consistent. (b) Proposition A.11 with (a). (c) *The global-schedule filter.* For a function machine F, read as a machine that reads the tape one-way, let F' answer a query φ by running the *global schedule* of F: stage s reads tape bit s, starts F on the s-th sentence and advances every started run by one step; certification events (a run halting with a label) are ordered by stage, ties by sentence index. When φ's event occurs, F' checks whether the set of labels certified up to and including it is consistent with B (decidable by hypothesis); if so it outputs φ's label, otherwise it diverges. The answer is a deterministic function of the bits read, so F' is a function machine, and $K(F') \le K(F) + c$. On a tape ω let $l_1, l_2, \dots$ be the labelled sentences in event order. F' outputs the label of l_i iff $\{l_1, \dots, l_i\}$ is consistent; consistency of these prefixes is monotone, so the outputs form an initial segment all of whose finite subsets are consistent, hence a consistent set, by compactness. So F' is consistent on every tape, and $F' = F$ on every tape on which F is consistent. Hence $\nu_{F'}(D) \ge \nu_F^{\mathrm{cons}}(D)$ for every D, and summing with (1) gives $M_F^{\mathrm{mw}} \ge 2^{-c} M_F^{\mathrm{cons}}$. The theory-machine version filters emissions in the same way. □

With two unary function symbols, consistency of finite sets is not decidable, so in (c) "monadic" must exclude function symbols.

Question A.22. Is $M_F^{\mathrm{cons}}/M_F^{\mathrm{mw}}$ bounded in general, for instance for $B = \mathsf{Q}$, or for $B = \emptyset$ and a language with a binary relation symbol?

A machine consistent on every tape must avoid inconsistency, which is only semi-decidable, so its support is an r.e. family of consistent data sets; a pathwise-conditioned machine can exploit the Π_1^0 event that nothing inconsistent ever happens. The lottery does not separate the two: the lottery machine is consistent on every tape (**??**).

A.5 Time bounds with free deduction

As in Appendix E.6 of the earlier report, the penalised coin-free hypotheses are programs of the machine U of Assumption A.5, so $\mathrm{time}_p(x) \ge |p|$, and $a' := \max(a, 2)$. Clocked strings, their weights $w(f)$ and $c_\perp$ are as in Section 1.5, with E the Elias-δ code of $|f| + 1$ followed by f; c_1 is the length of the fixed wrapper that runs a clocked string and diverges when it abstains. Simulations between U_0, U and multitape machines have polynomial overhead (standard; used in the same form as there).

Proposition A.23 (A Levin penalty costs a logarithm)**.** *Let $t(m) \ge m^{a^*}$ be time-constructible, m_0 the least length of a sentence, $W_A^{\mathrm{Kt},t}$ the mixture over all deciders p with weight $2^{-|p|}/c_p$, $c_p := \sup_\chi \mathrm{time}_p(\chi)/t(|\chi|)$ (Definition E.10 of the earlier report), and $W^1(D) := \sum_f 2^{-|f|}/(|f|+1)$ over the programs f of consistent labellers extending D. Under Assumption A.5, $W_A^{\mathrm{Kt},t}(\emptyset) \le t(m_0)$, and there is c with $2^{-c} W^1(D) \le W_A^{\mathrm{Kt},t}(D) \le 2^c t(m_0) W^1(D)$ for every D; so a labeller f pays $\log(|f| + 1) + O(1)$ bits beyond its unpenalised weight.*

Proof. This is Proposition E.13 of the earlier report. Since $\mathrm{time}_p \geq |p|$, $c_p \geq |p|/t(m_0)$, so p has weight at most $t(m_0)2^{-|p|}/|p|$, and the total is at most $t(m_0)$ by Kraft's inequality. For the lower bound, the padded decider of a consistent labeller f (as in the proof of Theorem A.9) has $c_d \leq \beta(|f|+1)$ for a constant β, by Lemma E.12(b) there and $(|\chi|+1)^{a'} \leq 2^{a'}t(|\chi|)$, so it has weight at least $2^{-|f|-2c}/(\beta(|f|+1))$; the map is injective. For the upper bound, map a compatible decider p to its proof-search labeller f_p, with $|p| \leq |f_p| \leq |p|+c$; the weight of p is at most $t(m_0)(c+2)2^c\,2^{-|f_p|}/(|f_p|+1)$, and $p \mapsto f_p$ is injective. □

Theorem A.24 (Domination lemma)**.** *Let X be a coin-free mixture and v_X a weight on consistent labellers such that $W_X(D) \geq 2^{-c_X}\sum_g v_X(g)$, the sum over the consistent labellers g extending D, and $W_X(\emptyset) \leq \Omega_X$. Let Y be a mixture with $W_Y(D) = \sum_h v_Y(h,D)$, the sum over the hypotheses h compatible with D, with an injective map $h \mapsto g_h$ to consistent labellers that compute the same partial map as h, with $v_Y(h,D) \leq k_Y\, v_X(g_h)$, and with $W_Y(\emptyset) \geq \kappa_Y > 0$. Then $\ell_X(D) \leq \ell_Y(D) + c_X + \log(k_Y\Omega_X/\kappa_Y)$ for every D.*

Proof. A labeller g_h computes the same partial map as h, so it extends D iff h is compatible with D. Hence $W_X(D) \geq 2^{-c_X}\sum_h v_X(g_h) \geq 2^{-c_X}k_Y^{-1}W_Y(D)$, and $\ell_X(D) = -\log W_X(D) + \log W_X(\emptyset) \leq -\log W_Y(D) + c_X + \log(k_Y\Omega_X) = \ell_Y(D) - \log W_Y(\emptyset) + c_X + \log(k_Y\Omega_X)$. □

The weaker hypothesis $\Gamma_{g_h} = \Gamma_h$ does not suffice: a labeller that rejects ψ and one that accepts $\neg\psi$ and abstains on ψ have the same Γ but differ on the datum $(\psi, 0)$.

Corollary A.25 (A one-sided equivalence)**.** *For every time-constructible τ there is c_τ such that, on every data set D, the losses of W_A and W_A^{dec} exceed that of $W^{\mathrm{cons},\tau}$ by at most c_τ, and the loss of M_A exceeds that of $M_F^{\mathrm{cons},\tau}$ by at most c_τ. Under Assumption A.5 the same holds for $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$, $t(m) \geq m^{a^*}$ time-constructible.*

Proof. Coin-free (Theorem 6.14(b) of the earlier report): apply Theorem A.24 with $X \in \{W_A, W_A^{\mathrm{dec}}, W_A^{\mathrm{chk},t}, W_A^{\mathrm{gen},t}\}$ and $Y = W^{\mathrm{cons},\tau}$. By Theorem 1.2 and, for the penalised versions under Assumption A.5, Theorem A.9(b), $W_X \geq 2^{-c}W^{\mathrm{cons}}$, and $W_X(\emptyset) \leq 1$ by Kraft's inequality. For a clocked string h let g_h be the wrapper that runs h^τ and diverges when it abstains, followed by $E(h)$; it computes the same partial map as h^τ, $|g_h| = |E(h)| + c_1$, and $h \mapsto g_h$ is injective. $W^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$, the weight of the always-abstaining string (B is consistent). So $c_\tau := c + c_1 + c_\perp$ works. One tape: the clocked function machine of a string f is a function machine whose V-program is a fixed wrapper followed by $E(f)$, injectively, so (1) gives $M_F^{\mathrm{cons}} \geq 2^{-c_1}M_F^{\mathrm{cons},\tau}$ on $D \neq \emptyset$; Theorem 1.3 gives $M_A \geq 2^{-c}M_F^{\mathrm{cons}}$; and $M_A(\emptyset) \leq 1$, $M_F^{\mathrm{cons},\tau}(\emptyset) \geq 2^{-c_\perp}$. □

Proof of Theorem 1.7. This is Theorem 6.15 of the earlier report; we recall the argument. By the MRDP theorem (Matiyasevich, 1970; Davis, 1973) there are polynomials E, E' with natural-number coefficients such that program e halts on x with output y iff $\mathbb{N} \models \exists \bar{u}\,(E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}))$; $\mathrm{Out}(e,x,y)$ is this formula (Lemma E.14(a) there), and $\mathrm{DET} := \forall e\,\forall x\,\forall y\,\forall y'\,\forall \bar{u}\,\forall \bar{u}'\,(E(e,x,y,\bar{u}) = E'(e,x,y,\bar{u}) \wedge E(e,x,y',\bar{u}') = E'(e,x,y',\bar{u}') \to y = y')$, a true Π_1 sentence. Let $H_0 := \mathsf{Q} \cup \{\mathrm{DET}\}$. *Construction.* Let $\mathcal{G}(e',n)$ be the program that, for $j = 1,\ldots,n$, forms $\sigma_j := \mathrm{Out}(\underline{e'}, \underline{j}, S0)$, asks P for approximations (a_0, a_1) within 2^{-j-3} of its pair on the history $((\sigma_1,\beta_1),\ldots,(\sigma_{j-1},\beta_{j-1}))$ and the sentence σ_j, and sets $\beta_j := 0$ if $a_0 \leq a_1$ and $\beta_j := 1$ otherwise; it outputs β_n. $\mathcal{G}$ is total, as P is. By Kleene's recursion theorem (standard) there is e with $e(n) = \mathcal{G}(e,n)$ for all n. The values β_j computed inside $\mathcal{G}(e,n)$ do not depend on n, so $e(j) = \beta_j \in \{0,1\}$ and $\sigma_j = s_j$; and $\mathbb{N} \models s_n$ iff $e(n) = 1$, so the label b_n is β_n, and P is queried exactly on the true histories D_{j-1}. *H_0 decides each s_n.* If $b_n = 1$, s_n is a true

existential equation, which Q proves (Lemma E.14(c) there). If $b_n = 0$, $\mathrm{Out}(\underline{e}, \underline{n}, 0)$ is one, so Q proves it, and with s_n and DET it gives $S0 = 0$, which Q refutes; so $H_0 \vdash \neg s_n$. *The loss of P.* With $q_b := q_b(D_{j-1}, s_j)$ and $\varepsilon := 2^{-j-3}$, the tie rule gives $q_{b_j} \le a_{b_j} + \varepsilon \le a_{1-b_j} + \varepsilon \le q_{1-b_j} + 2\varepsilon$, and with $q_0 + q_1 \le 1$ this gives $q_{b_j} \le \frac{1}{2} + 2^{-j-3}$. So $-\log q_{b_j} \ge 1 - \log(1 + 2^{-j-2})$, and $\sum_j \log(1+2^{-j-2}) \le \sum_j 2^{-j-2}/\ln 2 = 1/(4\ln 2) < 0.361$. *Axiom induction.* H_0 is true in $\mathbb{N}$ and B in an expansion of $\mathbb{N}$, so $B \cup H_0$ is consistent, and H_0 is compatible with every D_n. It is finite, so it has a decider running in linear time and an enumerator writing finitely many axioms; let $\omega_0 > 0$ be the least of their weights in W_A, $W_A^{\mathrm{chk},t}$ and $W_A^{\mathrm{gen},t}$. Then $W(D_n) \ge \omega_0$, $W(\emptyset) \le 1$, and the loss is at most $\log(1/\omega_0)$, whatever P is. *Clocked SOI.* The last claim concerns the sequence built against one predictor. Put $P^*(D) := (W^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$ and $q_b(D, \varphi) := P^*(D + (\varphi, b))/P^*(D)$. Then $q_0 + q_1 \le 1$, since $W^{\mathrm{all},\tau}$ is a semimeasure and the uniform part halves at each step; P^* is computable, since clocked compatibility is decidable and the weights have computable tails; $P^*(D) \ge 2^{-|D|-1} > 0$; and $W^{\mathrm{all},\tau}(\emptyset) = 1$, so $P^*(\emptyset) = 1$ and $\ell_{P^*}(D_n) = -\log P^*(D_n)$. On the sequence built against P^*, $W^{\mathrm{cons},\tau}(D_n) \le W^{\mathrm{all},\tau}(D_n) \le 2P^*(D_n)$, so $\ell_{W^{\mathrm{all},\tau}}(D_n) \ge n - 1.37$ and $\ell_{W^{\mathrm{cons},\tau}}(D_n) \ge \ell_{P^*}(D_n) - 1 + \log W^{\mathrm{cons},\tau}(\emptyset) \ge n - 1.37 - c_\perp$. □

Proposition A.26 (Incomparability with time-bounded predictors that may be inconsistent)**.** *On the data of Remark A.14, $W^{\mathrm{all},\tau}$ has bounded loss for every τ that covers the constant running time of "always 1", while every consistent inducer (SAI and consistent SOI, coin-free or one-tape) loses at least n bits in expectation; on a computable sequence (Theorem 1.7, applied to P^*), $W^{\mathrm{all},\tau}$ loses at least $n - 1.37$ while SAI loses $O(1)$. So $W^{\mathrm{all},\tau}$ and SAI are incomparable.*

Proof. Once τ covers its running time, the clocked string "always 1" is compatible with every prefix of the data of Remark A.14, so the loss of $W^{\mathrm{all},\tau}$ is at most $-\log w(\text{always 1})$. The other two claims are Remark A.14 and Theorem 1.7, the latter for the sequence built against P^*. □

Proof of Proposition 1.8. This is Proposition E.15(a), (b) of the earlier report. (a) $\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true iff e outputs 1 on n; if e halts on n with another output, $\neg\mathrm{Out}(\underline{e}, \underline{n}, S0)$ is true, as a program has at most one output. So every label of u is true in $\mathbb{N}$. (b) In Theorem 1.7 e is total with values in $\{0, 1\}$, so u labels every s_n correctly and extends every D_n; hence $W^{\mathrm{cons}}(D_n) \ge 2^{-|u|-c}$, where c pays for turning u into a V-program for a labeller index, and with $W^{\mathrm{cons}}(\emptyset) \le 1$ the loss is at most $|u| + c$. The same holds for $W^{\mathrm{all}} \ge W^{\mathrm{cons}}$. □

Remark A.27 (A time-hierarchy effect)*. Proof sketch.* We do not write out the version of the diagonal theorem that gives the compared predictors the same reuse of state, nor the dependence on the machine model. The claim is that under a time bound the sequence that defeats a time-bounded predictor is itself computable within a slightly larger bound. For the predictors S_p of Proposition E.15(c) of the earlier report, which spend time $p(n)\log n$ per step and reuse their state, the labels that defeat S_p are produced by a predictor with per-step time $O(p(n)\log n)$ and state reuse that predicts its own labels with certainty; so $S_{p'}$, for a slightly larger p' (for example $np(n)$), has bounded loss on the sequence that defeats S_p.

Proposition A.28 (Disguised labellers keep their weight)**.** *Let $t(m) \ge m^{a^*}$ be time-constructible and X be $W_A^{\mathrm{chk},t}$ or $W_A^{\mathrm{gen},t}$ (any form). For a class $\mathcal{C}$ of deductively closed theories let $X(\mathcal{C}; D)$ and $W_A^{\mathrm{dec}}(\mathcal{C}; D)$ be the weights of the hypotheses compatible with D whose theory is in $\mathcal{C}$. Under Assumption A.5 there is c with $2^{-c} W_A^{\mathrm{dec}}(\mathcal{C}; D) \le X(\mathcal{C}; D) \le 2^c W_A^{\mathrm{dec}}(\mathcal{C}; D)$ for all $\mathcal{C}$ and D. So the posterior odds of any two classes of positive weight, genuine and disguised alike, move by at most 2^{2c}, and a slow consistent labeller f keeps prior weight at least $2^{-|f|-c}$, as a padded Craig set.*

Proof. This is Proposition E.16 of the earlier report: the padded sets of Lemma E.12 there keep the theory and meet the penalty, which gives the lower bound; for the upper bound the penalised deciders are deciders with the same weights, and an enumerator q maps to the padded decider of "yes iff q writes the input", which keeps the theory. Divide by the total weights. □

Theorem A.29 (One tape and the companion paper's time-bounded mixtures)**.** *Let* $M_F^{\mathrm{cons},\tau} := \sum_f w(f)\nu_{F_f^\tau}^{\mathrm{cons}}$*, where* F_f^τ *runs* f *on* U_0 *with a tape-reading instruction for at most* $\tau(|\varphi|)$ *steps and abstains otherwise, and let* $M_F^{\mathrm{all},\tau}$ *drop the consistency restriction. Parts (b)–(d) assume that* $\mathcal{L} \supseteq \mathcal{L}_{\mathrm{ar}}$ *and that* B *is decidable and true in an expansion of* $\mathbb{N}$*.*

(a) $\ell_{M_A}(D) \le \ell_{M_F^{\mathrm{cons},\tau}}(D) + c_\tau$ *for every* D*.*

(b) On a computable sequence of arithmetic sentences, $\ell_{M_F^{\mathrm{all},\tau}} \ge n - 1.37$ *and* $\ell_{M_F^{\mathrm{cons},\tau}} \ge n - 1.37 - c_\perp$*, while* $\ell_{M_A} \le \log(1/\omega_0)$*.*

(c) The length-activated and round-activated mixtures of the companion paper (Definition 9.1 and Proposition 9.4 there), with computable weights whose tails are computable, are computable predictors, so each loses at least $n - 0.361$ *on a computable sequence on which SAI loses* $O(1)$*. Conversely, on the coin literals of Remark A.14, all labelled true, the round-activated mixture has bounded loss, while SAI and consistent SOI lose at least* n *bits in expectation. So the companion paper's time-bounded SOI and SAI are incomparable.*

(d) Unpenalised W^{cons} *and* M_F^{cons} *have loss at most* $|u| + c$ *on the diagonal sequences of (b) and (c).*

Proof. (a) is the one-tape part of the proof of Corollary A.25. (b) Each query of F_f^τ reads at most $\tau(|\varphi|)$ tape bits, so $\nu_{F_f^\tau}(D)$ is a finite union of cylinders, computable exactly; with computable tails, $M_F^{\mathrm{all},\tau}$ is computable, a semimeasure with value 1 at $\emptyset$, and $P(D) := (M_F^{\mathrm{all},\tau}(D) + 2^{-|D|})/2$ is a computable predictor; run Theorem 1.7 against P, as in its last step. The theory machine that emits H_0 is compatible with every D_n on every tape. (c) Theorem 1.7 needs only computability of the predictor. The converse is Proposition 9.4 of the companion paper with the constant comparator "label true with probability 1", which is in its class $\mathcal{C}_t$, and Proposition E.1 of the earlier report, applied tape by tape (Remark A.14). (d) Proposition 1.8, with (1) for M_F^{cons}. On the coin literals of (c), by contrast, every consistent inducer, W^{cons} and M_F^{cons} included, loses at least n bits in expectation. □

A.6 Charged deduction

We use the setting of Definition E.18 of the earlier report: $\mathcal{L}$ is finite, B is decidable in polynomial time, derivations are in the Hilbert calculus K of that report, with sizes counting symbols, and a hypothesis is a program p that decides an axiom set A_p in time $O(t)$. Each datum φ must have a derivation of φ^b from $B \cup A_p$ whose size is at most $s(|\varphi|)$ ($W_A[t,s]$); in which every axiom instance, logical ones included, has at most $h(|\varphi|)$ symbols (reading (a1), $W_A^{a1}[t,h]$), or every nonlogical instance does (reading (a2), $W_A^{a2}[t,h]$); or whose *material*, the total size of the distinct instances used, is at most $g(|\varphi|)$ (reading (b), $W_A^b[t,g]$), or whose nonlogical material is (reading (b2), $W_A^{b2}[t,g]$). *Certificate* SOI, $W^{\mathrm{cert}}[t',s']$, weighs the consistent labellers $f_{\mathcal{V}}$ defined by verifiers $\mathcal{V}$ running in time $O(t')$: $f_{\mathcal{V}}(\varphi)$ is 1 (resp. 0) if some certificate of length at most $s'(|\varphi|)$ makes $\mathcal{V}$ accept (resp. reject). All charged results assume the renaming condition of that report; without it the index of a single parameter, which costs one symbol, could carry a certificate of any length (Remark A.34).

Assumption A.30 ((F1): renaming closure)**.** For every hypothesis p of a charged reading, $B \cup A_p$ is closed under permutations of the parameters.

Theorem A.31 (Charged deduction is certificate SOI)**.** *Under Assumption A.30, with time-constructible nondecreasing budgets: (a) in a normal derivation of φ every line sits at its own node of an axiom instance used or of φ, up to abstracting parameters; so its size is at most $(M + |\varphi|)(M + |\varphi| + 1)/2$, with M its material, and the factor $\frac{1}{2}$ is attained; (b) with $G(m) := (g+m+1)(g+m+2)/2$, $t^+(m) := m^{k_0}(t(m)+1)$ and $t'^{\#}(m) := m^{k_0}+t'(m)$, for constants k_0, β and c: $W_A[t,g] \le W_A^b[t,g] \le W_A[t,G]$ hypothesis by hypothesis; $W_A^b[t,g] \le 2^c\, W^{\mathrm{cert}}[t^+, \beta G \log(G+2)]$; and $W^{\mathrm{cert}}[t',s'] \le 2^c$ times each of $W_A^b[t'^{\#}, 2s'+m+31]$, $W_A^{a1}[t'^{\#}, s'+m+16]$ and $W_A[t'^{\#}, 2s'+2m+31]$.*

Its classical core is the correspondence of propositional proof systems with NP (Cook and Reckhow, 1979): a derivation is a certificate; conversely, a certificate ζ becomes the axiom $\theta_\zeta \to \varphi^b$, where θ_ζ is a tautology of size $|\zeta| + 14$ that encodes ζ.

Proof. This is Theorem 6.16 of the earlier report, with Lemma E.19 and Proposition E.21 there, whose budgets are time-constructible and nondecreasing. Part (a) is the subformula lemma (Lemma E.19(b), (d) there): in a normal derivation every line sits at its own formula node of an instance used or of the conclusion, up to abstracting the parameters of a chain of Gen steps, and a family with k vacuous Gen steps attains the factor $\frac{1}{2}$. Part (b): a derivation of size at most g has material at most g, and by (a) material at most g gives size at most G; this is the first chain, hypothesis by hypothesis. Under Assumption A.30 the parameters of a derivation of size at most G can be renamed to $p_0, \dots, p_{G-1}$, so the derivation is a certificate of at most $\beta G \log(G+2)$ bits that a verifier checks in time t^+ (membership in A_p in time t per line); this is the second inequality. For the third, let Z be a closed tautology of size 4 and C_ζ a sentence of size $|\zeta| + 4$ that encodes the certificate ζ, and put $\theta_\zeta := Z \to (C_\zeta \to Z)$, a tautology of size $|\zeta| + 14$. The axiom set $\{\theta_\zeta \to \varphi^b : \zeta$ a certificate on which $\mathcal{V}$ accepts $(b = 1)$ or rejects $(b = 0)$ $\varphi\}$ is decidable in time $t'^{\#}$, satisfies Assumption A.30, and has the same consequences as $\Gamma_{f_\mathcal{V}}$. The derivation θ_ζ, $\theta_\zeta \to \varphi^b$, φ^b has material $2|\zeta| + 29 + |\varphi^b|$, largest instance $|\zeta| + 15 + |\varphi^b|$ and size $2|\zeta| + 29 + 2|\varphi^b|$, with $|\varphi^b| \le |\varphi| + 1$; these give the three stated budgets. All maps are injective and add a constant to the length. □

Corollary A.32 (Fit classes)**.** *Under Assumption A.30, on literal sequences D^X, at polynomial budgets whose degrees are charged in the prior, with standard simulation overheads:*

(i) reading (b) fits exactly the $X \in \mathsf{NP} \cap \mathsf{coNP}$; it beats polynomial-time consistent SOI, $W^{\mathrm{cons,poly}}$, on some sequence if $\mathsf{NP} \cap \mathsf{coNP} \neq \mathsf{P}$, and fits the same sequences if $\mathsf{P} = \mathsf{NP}$; with degrees uncharged, $W^{\mathrm{cons,poly}}$ fails to dominate it, unconditionally; and, under Assumption A.5, it beats $W^{\mathrm{cons},\tau}$ at a budget a hierarchy gap above τ;

(ii) reading (a1) fits exactly the $X \in \mathsf{EXPTIME}$; at any computable budgets some decidable X defeats it; and every predictor computable in time polynomial in the number of data loses at least $n - 0.361$ on a sequence on which (a1) at a linear budget loses $O(1)$;

(iii) readings (a2) and (b2) fit every decidable X, by a finite set of universally closed Horn sentences.

Proof. This is Corollary 6.17(a)–(c) of the earlier report, with its proofs there. (ii) uses $\mathsf{APSPACE} = \mathsf{EXPTIME}$ (Chandra et al., 1981) through Proposition E.23 there, and the comparisons use the time hierarchy theorems (standard; cf. Hartmanis and Stearns, 1965, and Seiferas et al., 1978 for nondeterministic time). □

Proposition A.33 (Soft charges)**.** *Let a hypothesis pay a factor $2^{-\kappa\ell}$ per datum, ℓ the bit length of a shortest derivation of the datum (soft SAI), and let a verifier pay likewise for its shortest certificate (soft certificate SOI). The two correspond only up to a constant factor in the rate κ; no pair of rates gives two-sided constant regret; and every soft loss diverges on every infinite sequence, so there are no soft fit classes.*

Proof. Proposition E.26 of the earlier report, which gives the explicit rates. □

Remark A.34 (The role of the renaming condition)*.* Without Assumption A.30 the index of one parameter, which costs one symbol, can carry a certificate of any length (Proposition E.27 of the earlier report). The coarse statements (Theorem 6.9 there, and the inclusion of the fit class of reading (b) in $\mathsf{NP} \cap \mathsf{coNP}$) survive in the reading that counts symbols, by truncating parameter indices (Lemma E.28 there); the fine bounds ($2^{O(h \log h)}$ for (a1), certificate lengths) are not claimed without it.

Remark A.35 (One tape)*. Proof sketch.* We do not write out the bookkeeping of a verifier that reads a tape, which needs Assumption A.6-style accounting. The maps of Theorem A.31(b) act tape by tape: a random decider with time $O(t)$ on every tape gives a verifier that reads the tape to answer membership queries, and certificate axioms $\theta_\zeta \to \varphi^b$ can be emitted from a random verifier. So a one-tape form of the sandwich should hold with the same budgets.

A.7 Presburger arithmetic

Remark A.36 (Lengths)*.* (i) The length $|\varphi|$ of a sentence is its symbol count: every occurrence of a connective, quantifier, nonlogical symbol, parenthesis or variable counts 1, whatever the variable's index. This is the size measure of the calculus K of the earlier report, in which a variable or parameter counts one symbol. Machines read φ as a bit string of length $b(\varphi) \le \kappa_0 |\varphi| \log(|\varphi| + 2)$, since a sentence with m symbols has at most m distinct variables, which can be renamed $x_1, \dots, x_m$ and written in binary. Clocks τ, derivation budgets s, g, h and the length m in the lower bounds are functions of $|\varphi|$. (ii) If a clock were a function of the bit length b, which can exceed the symbol count by a factor $\log m$, a clock $2^{2^{c'b}}$ could exceed every $2^{2^{cm}}$, and the contradictions below would fail. For clocks in bit length the results hold with $\tau(b) \le 2^{2^{c'b/\log(b+2)}}$ in place of $\tau(m) \le 2^{2^{c'm}}$, with $c' < c/\kappa_0$ (and likewise for the singly exponential clocks), since $b/\log(b+2) \le \kappa_0 |\varphi|$. (iii) *Proviso:* Fischer and Rabin's length counts every symbol occurrence, variables included, at least once, and their nonlogical symbols are among $0, 1, +, =, <$. Under the proviso every sentence of their length n has an equivalent $\mathcal{L}_{\mathrm{Pr}}$-sentence with at most κn symbols, computable in polynomial time (rewrite $x < y$ as $\exists z\,(x + z + 1 = y)$ with z fresh, and copy every other symbol). *Padding:* every $\mathcal{L}_{\mathrm{Pr}}$-sentence ψ has an equivalent sentence with exactly m symbols for every $m \ge |\psi| + c_0$, computable in polynomial time, because there are two equivalence-preserving paddings with coprime increments (in the size measure of K, $\psi \wedge 0 = 0$ adds 4 and $\psi \wedge \neg 0 = 1$ adds 5; every standard syntax has such a pair). *Consequence:* let Q be a decision procedure for PrA in our syntax, deterministic or nondeterministic. For $j = 0, \dots, \kappa - 1$ let P_j decide their sentences of length n by translating, padding to exactly $\kappa n + c_0 + j$ symbols and running Q. Each P_j is a decision procedure in their syntax, with the same reading of nondeterministic time as Q; so by their theorem, for all large n, Q takes more than $2^{2^{c_{\mathrm{FR}} n}} - \mathrm{poly}(n)$ steps on some sentence with exactly $\kappa n + c_0 + j$ symbols. As j runs through all residues, for all large m, Q takes more than $2^{2^{cm}}$ steps on some sentence with exactly m symbols, for every $c < c_{\mathrm{FR}}/\kappa$; c denotes such a constant. The exact-length form is what the "for all large m" arguments below use.

We use two known results. ($\mathrm{F_P}$) Presburger's axioms for successor and addition, with the induction schema for $\mathcal{L}_{\mathrm{Pr}}$-formulas, form a decidable, consistent and complete axiomatisation of PrA (Presburger, 1929), with membership decidable in polynomial time. ($\mathrm{F_{FR}}$) The bound of Fischer and Rabin (1974), in the form of Remark A.36(iii). Its statement does not say how the time of a nondeterministic procedure is measured. We distinguish two readings: (max) the time is the maximum over the computation paths, all of which halt, every output being correct and some path giving an output; (short) the time on a sentence is the length of the shortest computation that outputs the correct answer. (short) implies (max). Where only (max) is needed nothing is assumed; the one place that uses (short) says so.

Proposition A.37 (Presburger arithmetic separates SAI from clocked SOI)**.** *(a) W_A, W_A^{dec} and M_A, and the penalised $W_A^{\mathrm{chk},t}$, $W_A^{\mathrm{gen},t}$ and $M_A^{\mathrm{chk},t}$ for a large enough polynomial t, have bounded loss on D^{Pres}. (b) If τ is time-constructible and $\tau(m) \le 2^{2^{c'm}}$ for large m, with $c' < c$, the losses of $W^{\mathrm{cons},\tau}$ and $W^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity. (c) If $\tau(m) \le 2^{c'm}$ for large m, with $c' < c$, the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.*

Proof. (a) The decider of Presburger's axioms is compatible with every prefix of D^{Pres} (the axiomatisation is complete and consistent), runs in polynomial time and is a deterministic theory machine; with an enumerator of the same axioms it meets the axiom-side penalties for a large enough polynomial t. The losses are at most $-\log$ of its weight. (b) In a clocked mixture compatibility is inherited by prefixes, so the compatible strings form a decreasing family and bounded loss gives one τ-clocked f that labels every sentence correctly (Lemma E.11 of the earlier report). By the standard simulation overhead it yields a multitape decision procedure for PrA that runs in time $q(C_f\tau(m)+b)$, with q a fixed polynomial and $b \le \kappa_0 m\log(m+2)$ the input's bit length. When $c' < c$ this is at most $2^{2^{cm}}$ for large m, which contradicts $\mathrm{F_{FR}}$ in the deterministic case. (c) Write $M := \sum_f w(f)\nu_f$ for either one-tape mixture, each $\nu_f(D_n^{\mathrm{Pres}})$ being non-increasing in n. Bounded loss gives $\inf_n M(D_n^{\mathrm{Pres}}) > 0$; by dominated convergence some f has $\lim_n \nu_f(D_n^{\mathrm{Pres}}) > 0$, which by continuity from above is $\mathbb{U}(\mathcal{G}_f)$ for the set $\mathcal{G}_f$ of tapes on which F_f^τ labels every sentence correctly (and consistently, for the consistent version). By the Lebesgue density theorem (**??**) there is w_0 with $\mathbb{U}(\mathcal{G}_f \cap [w_0]) > \frac{3}{4}\,2^{-|w_0|}$, so for every sentence the tapes in $[w_0]$ on which F_f^τ answers correctly have conditional measure greater than $\frac{3}{4}$. A query of length m reads at most $\tau(m)$ tape bits, so the procedure "for every v of length $\max(0,\tau(m)-|w_0|)$ run F_f^τ with tape w_0v and output the majority answer" (w_0 hard-coded) decides PrA deterministically within $2^{\tau(m)}q(C_f\tau(m)+b) \le 2^{2^{cm}}$ steps for large m: a contradiction. The majority vote over the tape completions at a density point is the technique of de Leeuw et al. (1956); the argument applies to every mixture of machines whose tape use per query is bounded by τ. □

A computation with illustrative values of the constants (code in the accompanying repository; c is not known explicitly) gives the lengths from which these contradictions apply: for $c = 1$ and $c' = 0.9$, from about 11–26 symbols in (b) and 6–9 in (c). The separations are asymptotic: below these lengths a clocked hypothesis may label every short sentence correctly.

Assumption A.38 ((B*): a Σ_2-time lower bound for Presburger arithmetic)**.** There is $c_B > 0$ such that, for all large m, no alternating machine with at most two alternations (a Σ_2 machine) decides PrA within $2^{2^{c_B m}}$ steps on all sentences with m symbols.

Assumption A.39 ((SGL): time-bounded $\mathsf{BPP} \subseteq \Sigma_2$)**.** For time-constructible $T(m) \ge m$ and large T, $\mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$.

We do not prove either assumption. PrA is complete for alternating doubly exponential time with linearly many alternations (cf. Berman, 1980); (B*) would follow from the hardness part if the reductions were of linear size, and we do not assume that they are. If they are only of polynomial size, a bound $2^{2^{c_B m^\epsilon}}$ follows for some $\epsilon \in (0,1]$, and the clock condition below must be weakened to $\tau(m) \le 2^{2^{c' m^\epsilon}}$. (SGL) is the time-bounded form of $\mathsf{BPP} \subseteq \Sigma_2^p$ (Sipser, 1983; Lautemann, 1983): amplifying by $O(\log T)$ repetitions to $R = O(T \log T)$ random bits and error below $1/(2R)$, a Σ_2 machine guesses R shifts of R bits, universally chooses R bits, and runs the amplified machine R times, in time $O(T^2 \log^2 T)$.

Proposition A.40 (One tape, doubly exponential clocks)**.** *Under Assumptions A.38 and A.39, if $\tau(m) \le 2^{2^{c' m}}$ for large m with $c' < c_B$, then the losses of $M_F^{\mathrm{cons},\tau}$ and $M_F^{\mathrm{all},\tau}$ on D_n^{Pres} tend to infinity.*

Proof. As in the proof of Proposition A.37(c), obtain w_0 such that on $[w_0]$ the conditional probability of a correct answer exceeds $\frac{3}{4}$ for every sentence. The algorithm "draw v of length $\tau(m)$ at random, run F_f^τ with tape $w_0 v$, and output its answer" errs with probability below $\frac{1}{4}$ on every sentence and runs in time $T(m) := q(C_f \tau(m) + b)$, so $\mathrm{PrA} \in \mathsf{BPTIME}(T) \subseteq \Sigma_2\text{-}\mathsf{TIME}(T^3)$ by Assumption A.39. Since $T(m)^3 \le 2^{2^{c_B m}}$ for large m when $c' < c_B$, this contradicts Assumption A.38. □

A nondeterministic lower bound alone does not finish the doubly exponential case by this route: separating $\mathsf{NTIME}(2^{2^n})$ from Σ_2-time below it is not known in general.

Proposition A.41 (Charged deduction on Presburger data)**.** *Let p be a hypothesis of a charged reading over $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$, deciding A_p in polynomial time and satisfying Assumption A.30, and compatible with every prefix of D^{Pres}. Let $c'' < c$. (a) Size reading, budget s: $s(m) > 2^{2^{c'' m}}$ for infinitely many m, and for all large m if s is time-constructible. (b) Material reading (b), budget g: the same with g in place of s. (c) Reading (a1), with h nondecreasing and time-constructible: $h(m) > 2^{c'' m}$ for all large m.*

A mixture of such hypotheses has bounded loss on D^{Pres} only if one of them is compatible with every prefix.

Proof. Since compatibility is inherited by prefixes, it suffices to treat one hypothesis p, deciding A_p in time $O(m^{k_p})$ and compatible with every prefix of D^{Pres}. Then $B \cup A_p = A_p$ is consistent and proves every true sentence, so it proves exactly PrA. (a) *Infinitely often.* Fix $c'' < c$ and suppose $s(m) \le \tau''(m) := 2^{2^{c'' m}}$ for all $m \ge m_1$; τ'' is time-constructible. The nondeterministic procedure N: on a sentence φ with $m < m_1$ symbols, look φ up in a table; otherwise compute $\tau''(m)$, guess a derivation of φ or of $\neg\varphi$ of size at most $\tau''(m)$ whose parameter numbers have at most $K + 1 + \lceil \log(\tau''(m)+1) \rceil$ bits, $K = O(\tau''(m)^{k_p})$ (if a derivation of size at most $\tau''(m)$ exists, such a one exists, by Lemma E.28 of the earlier report), check it as in the proof of Theorem 6.9 there (each line a logical axiom, by matching, a member of A_p, in time $O(\tau''^{k_p})$, or a step of modus ponens or generalisation), and output the label it proves; a path whose guess is not a valid derivation outputs nothing. Every output is correct, every sentence has a path with an output, and every path halts within $\tau''(m)^{c_k}$ steps, c_k depending only on k_p and the checking overhead of K. Since $\tau''(m)^{c_k} = 2^{c_k 2^{c'' m}} \le 2^{2^{cm}}$ once $m \ge \log(c_k)/(c - c'')$, N contradicts $\mathrm{F_{FR}}$ (nondeterministic case, reading (max)). So $s(m) > 2^{2^{c'' m}}$ for infinitely many m. *For all large m, s time-constructible.* Suppose $s(m) \le \tau''(m)$ for infinitely many m, the *good* lengths. The procedure N'': on a sentence

with m symbols, compute $s(m)$ with a clock of $\tau''(m)$ steps; if the computation finishes with $s(m) \le \tau''(m)$, run N's guess-and-check with the budget $s(m)$; otherwise run a deterministic decision procedure for PrA, which exists by $\mathrm{F_P}$. Every output is correct and every path halts, and at each good length every path halts within $O(\tau''(m)) + \tau''(m)^{c_k} \le 2^{2^{cm}}$ steps beyond the thresholds. By $\mathrm{F_{FR}}$ (reading (max)), for all large m some sentence with m symbols takes N'' more than $2^{2^{cm}}$ steps; a good length beyond the thresholds gives a contradiction. (For an arbitrary s the same conclusion follows if $\mathrm{F_{FR}}$ holds in reading (short): let a path whose guess is not a valid derivation run the deterministic procedure; at each good length every sentence then has a correct output within $\tau''(m)^{c_k} + O(1)$ steps.) (b) By the subformula lemma (Theorem A.31(a)), material at most g gives a derivation of size at most $(g+m)(g+m+1)/2$. If $g(m) \le 2^{2^{c''m}}$ for all large m, this is at most $2^{2^{c'''m}}$ for large m, for any $c''' \in (c'', c)$, against (a) with c'''. The form for all large m transfers in the same way. (c) Proposition E.23(a) of the earlier report gives the labeller g_p: with $H(\varphi) := \max(h(|\varphi|), |\varphi| + 1)$ and Der_H the formulas with a derivation from $B \cup A_p$ all of whose lines have size at most H, it accepts φ if $\varphi \in \mathrm{Der}_{H(\varphi)}$, rejects it if $\neg\varphi \in \mathrm{Der}_{H(\varphi)}$ and $\varphi \notin \mathrm{Der}_{H(\varphi)}$, and abstains otherwise. It runs in time $2^{O(H \log H)}(1+t(H))^{O(1)}$, by closing the at most $2^{O(H \log H)}$ canonical formulas of size at most H under the rules, using Assumption A.30; and it labels correctly every datum of every D with which p is (a1)-compatible, since a normal (a1)-derivation of φ^b has every line of size at most $\max(h(|\varphi|), |\varphi^b|) \le H(\varphi)$ (the subformula lemma) and $B \cup A_p$ is consistent. Nothing in this argument uses the form of φ. On D^{Pres}, g_p therefore decides PrA deterministically (it never abstains, since every sentence is a datum) in time $T(m) \le 2^{C_a H \log H}$ with $H = \max(h(m), m+1)$ and C_a depending on k_p (computing $h(m)$ costs $O(h(m))$). By $\mathrm{F_{FR}}$, $T(m) > 2^{2^{cm}}$ for all large m, so $C_a H \log H > 2^{cm}$, hence $H > 2^{cm}/(C_a cm) > 2^{c''m}$ for all large m; as $m + 1 < 2^{c''m}$ for large m, $H = h(m)$. □

Question A.42. (i) Under reading (a1), does every hypothesis compatible with every prefix of D^{Pres} need doubly exponential instance budgets h? (ii) Does some hypothesis of reading (a1) in $\mathcal{L}_{\mathrm{Pr}}$ fit D^{Pres} at exponential budgets? (iii) What do readings (a2) and (b2) need on D^{Pres}, in either direction?

(i) would follow from a deterministic lower bound PrA $\notin \mathsf{DTIME}(2^{2^{2^{c_3 m}}})$ for some $c_3 > 0$, by the argument of Proposition A.41(c), which would then give $h(m) > 2^{2^{c_3' m}}$ infinitely often for every $c_3' < c_3$; no such bound is known to us. For (ii), Proposition E.23(b) of the earlier report simulates alternating space-bounded machines, but it needs the literal language with two extra binary relation symbols, which $\mathcal{L}_{\mathrm{Pr}}$ with $B = \emptyset$ lacks. For (iii), logical instances are free under (a2) and (b2), so neither the guess-and-check route nor the enumeration route gives a bound, and Corollary A.32(iii) concerns literal data in the literal language, not D^{Pres}.

Each charged reading matches one version of SOI. Once deduction is charged, the budget SAI needs on D^{Pres} is that of the matching version of SOI: under the size and material readings SAI is certificate SOI (Theorem A.31), and the nondeterministic Fischer–Rabin bound forces doubly exponential budgets on both; under (a1) SAI behaves like deterministic SOI with an exponential clock (Proposition E.23 and Corollary 6.17(b) of the earlier report; the converse direction is proved there for literal data only, so for sentence data we offer this as a reading), and the deterministic bound forces exponential instance sizes, which the clock turns into doubly exponential time. So on a natural theory, as on the diagonal sequences, the separation from clocked SOI is the free deduction, for every reading in which the derivation is charged and the conclusion is proved; for (a2) and (b2) the question is open.

A.8 The circuit form of the sandwich

The sandwich of Theorem A.31 has a routine circuit form, which connects finite axiom sets with the circuit Bayes theorem of the companion paper. Fix $d \geq 1$; the data are the literals $\varphi_x := R(t_x)$, $x \in \{0,1\}^d$, labelled by a partial $f : \{0,1\}^d \rightharpoonup \{0,1\}$.

A *verifier circuit* $\mathcal{V}$ of size $|\mathcal{V}|$ is a list of $|\mathcal{V}|$ two-input gates (any of the 16 Boolean functions) over the inputs $x_1, \ldots, x_d$, certificate bits $z_1, \ldots, z_k$ (without loss of generality $k \leq 2|\mathcal{V}|$), the constants and earlier gates; its last two gates are the outputs a (accept) and r (reject). It is *consistent* if no x has certificates ζ, ζ' with $\mathsf{a}(x,\zeta) = 1$ and $\mathsf{r}(x,\zeta') = 1$, and it defines $f_{\mathcal{V}}(x) := 1$ if some ζ has $\mathsf{a}(x,\zeta) = 1$, 0 if some ζ has $\mathsf{r}(x,\zeta) = 1$, and undefined otherwise. $\mathrm{vs}(f)$ is the least size of a consistent verifier circuit with $f_{\mathcal{V}} \supseteq f$; it is at most the circuit size of f in the sense of Definition 9.7 of the companion paper plus a constant. The *fixed language* is $\mathcal{L}_{\mathrm{circ}} := \{\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, \mathrm{Dig}, \mathrm{Gate}, R\}$, with Dig binary and Gate ternary. Index terms $\mathbf{n}_i$ and gate terms $\mathbf{g}_j$ are the word terms $t_{\beta(i)}$, $t_{\beta(j)}$, with $\beta(i)$ the binary expansion of $i+1$ without its leading 1; so $|\mathbf{n}_i| = O(\log i)$, the terms of each family are pairwise distinct, and the language does not grow with the circuit. $\mathrm{ax}_s(f)$ is the least total size of a finite set T of sentences of $\mathcal{L}_{\mathrm{circ}}$ (or of any fixed finite language containing $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1, R$) with $T \cup \Gamma_f$ consistent such that every datum $\varphi_x^{f(x)}$, $x \in \mathrm{dom}\, f$, has a K-derivation from T of size at most s.

The translation $T_{\mathcal{V}}$ (all sentences universally closed; $\leftrightarrow$ abbreviates $\neg((A \to B) \to \neg(B \to A))$). Digits: $\forall y\, \mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_1 y)$, $\forall y\, \neg\mathrm{Dig}(\mathbf{n}_0, \mathsf{s}_0 y)$, and $\forall y\, (\mathrm{Dig}(\mathbf{n}_{i+1}, \mathsf{s}_a y) \leftrightarrow \mathrm{Dig}(\mathbf{n}_i, y))$ for $i < \max(d,k) - 1$ and $a \in \{0,1\}$; so $\mathrm{Dig}(\mathbf{n}_i, t_w)$ holds iff the $(i+1)$-st symbol of w is 1. Gates: a gate j that reads x_i (resp. z_i) has the axiom $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{Dig}(\mathbf{n}_{i-1}, x))$ (resp. with z in place of x), and an internal gate j with operation op on gates a, b has $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_j, x, z) \leftrightarrow \mathrm{op}(\mathrm{Gate}(\mathbf{g}_a, x, z), \mathrm{Gate}(\mathbf{g}_b, x, z)))$. Outputs: $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{a}}, x, z) \to R(x))$ and $\forall x\, \forall z\, (\mathrm{Gate}(\mathbf{g}_{\mathsf{r}}, x, z) \to \neg R(x))$.

Theorem A.43 (The circuit sandwich)**.** *(i) For every consistent verifier circuit $\mathcal{V}$ with k certificate bits, $T_{\mathcal{V}}$ has total size at most $C_1(|\mathcal{V}| + d + k)\log(|\mathcal{V}| + d + k + 2)$, $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$ is consistent, and every $x \in \mathrm{dom}\, f_{\mathcal{V}}$ has a derivation of $\varphi_x^{f_{\mathcal{V}}(x)}$ from $T_{\mathcal{V}}$ of size at most $s_{\mathcal{V}} := C_2(|\mathcal{V}|(\log(|\mathcal{V}| + 2) + d + k) + (d+k)^3)$, with absolute constants C_1, C_2. Hence, using $k \leq 2|\mathcal{V}|$, $\mathrm{ax}_s(f) \leq C_1'(\mathrm{vs}(f) + d)\log(\mathrm{vs}(f) + d + 2)$ for $s = C_2'(\mathrm{vs}(f)(\log \mathrm{vs}(f) + d) + (d + \mathrm{vs}(f))^3)$. (ii) Conversely, $\mathrm{vs}(f) \leq P(\mathrm{ax}_s(f), s, d)$ for a fixed polynomial P that depends only on the language and K.*

This is the routine circuit form of the sandwich, not a separate result of independent interest; we include it because it carries the circuit Bayes theorem of the companion paper over to verifier circuits (Corollary A.44).

Proof. (i) *Size.* There are $O(d+k)$ digit axioms of size $O(\log(d+k))$, one axiom of size $O(\log(|\mathcal{V}| + d + k))$ per gate (two or three gate terms and at most two index terms), and two output axioms. *Consistency.* Take the free term algebra on $\mathsf{e}, \mathsf{s}_0, \mathsf{s}_1$, with equality the identity; every element is a word term. Read off the bits of an element y (the symbols of the word it spells, 0 beyond its length). Let $\mathrm{Dig}(t, y)$ hold iff $t = \mathbf{n}_i$ for some $i < \max(d,k)$ and bit i of y is 1; let $\mathrm{Gate}(t, x, z)$ hold iff $t = \mathbf{g}_j$ and gate j of $\mathcal{V}$ outputs 1 on the first d bits of x and the first k bits of z; and let $R(x)$ hold iff some ζ has $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$. The digit and gate axioms hold by construction. The first output axiom holds by the definition of R; the second holds because $\mathcal{V}$ is consistent: $\mathsf{r}(\mathrm{bits}_d(x), z) = 1$ excludes $\mathsf{a}(\mathrm{bits}_d(x), \zeta) = 1$ for every ζ. On word terms of length d, $R(t_x)$ holds iff $f_{\mathcal{V}}(x) = 1$, and $f_{\mathcal{V}}(x) = 0$ forces $\neg R(t_x)$; so the structure satisfies $T_{\mathcal{V}} \cup \Gamma_{f_{\mathcal{V}}}$. *Derivations.* For $x \in \mathrm{dom}\, f_{\mathcal{V}}$ take a certificate ζ for its label. Instantiate every gate axiom with $x := t_x$, $z := t_\zeta$ (two instances of the quantifier axiom and two modus ponens steps per axiom, each line of size $O(\log|\mathcal{V}| + d + k)$).

Derive each needed digit, $\mathrm{Dig}(\mathbf{n}_i, t_w)$ or its negation, from an instance of the base axiom and i instances of the successor axioms (each of size $O(d+k)$), substituting at each step a fixed propositional derivation for biconditional elimination: $O((d+k)^2)$ per digit, $O((d+k)^3)$ in all. Then, gate by gate in order, derive $\mathrm{Gate}(\mathbf{g}_j, t_x, t_\zeta)$ or its negation from the ground biconditional and the literals of its inputs, by a substitution instance of a fixed propositional derivation (the propositional fragment of K is complete, and substitution instances of derivations are derivations of size linear in the substituted formulas, as in the proof of Theorem 6.16 of the earlier report). Finish with the output instance and modus ponens. The total is $O(|\mathcal{V}|(\log|\mathcal{V}|+d+k)+(d+k)^3)$. (ii) A certificate is the code of a derivation of size at most s. Since T is a set of sentences, Assumption A.30 holds, and the parameters can be renamed to $p_0, \dots, p_{s-1}$, so the code has $O(s \log s)$ bits (Proposition E.21(b) of the earlier report). The checker, with T hard-wired, verifies each line (logical axioms by matching, quantifier axioms by first-order matching, members of T by comparison, modus ponens, generalisation) and the last line against φ_x or $\neg\varphi_x$, in time polynomial in $s+|T|+d$; it accepts if the code is a derivation of φ_x and rejects if it is one of $\neg\varphi_x$. Simulating it by circuits (standard; size $O(T \log T)$ for time T, Pippenger and Fischer, 1979) gives a verifier circuit of size $P(|T|, s, d)$. It is consistent because T is consistent, and $f_{\mathcal{V}} \supseteq f$ because every datum has a derivation within s and T proves no wrong label, $T \cup \Gamma_f$ being consistent. □

A computation (code in the accompanying repository) checks the translation on random consistent verifier circuits with $d \in \{2,3\}$: every axiom of $T_{\mathcal{V}}$ is true in the intended structure on a finite fragment, forward propagation through the ground instances derives the right label for every $x \in \operatorname{dom} f_{\mathcal{V}}$, every corruption of one gate that changes $f_{\mathcal{V}}$ is detected, and $|T_{\mathcal{V}}|$ stays between 7.9 and 9.0 times $(|\mathcal{V}|+d)\log(|\mathcal{V}|+d)$ for sizes from 33 to 1099 gates.

Corollary A.44 (Bayes over verifier circuits)**.** *Let W_{ver} be the mixture over consistent verifier circuits with prior $w_{\mathrm{ver}}(\mathcal{V}) := 2^{-|\mathcal{V}|}\,(2|\mathcal{V}|+1)^{-1}\,\Lambda^{-1}_{|\mathcal{V}|,k}$, where $k \le 2|\mathcal{V}|$ is uniform and $\Lambda_{j,k} := \prod_{i=1}^{j} 16(d+k+1+i)^2$ is the number of gate lists of size j. For a nonempty $J \subseteq \{0,1\}^d$, an ordering of J and the labels of f, let $L := -\log W_{\mathrm{ver}}(f|_J)$, $\ell_\emptyset := -\log W_{\mathrm{ver}}(\emptyset)$, and $\mathrm{Err} := \sum_i \epsilon_i$, with ϵ_i the posterior mass at step i that is wrong* or undefined *at x_i. Then: (a) some consistent verifier circuit of size at most $\lceil L \rceil$ agrees with f on J, and $L \le \mathrm{vs}_J + \log(2\mathrm{vs}_J+1) + \log \Lambda_{\mathrm{vs}_J, 2\mathrm{vs}_J} = O(\mathrm{vs}_J \log(\mathrm{vs}_J + d))$, with $\mathrm{vs}_J := \mathrm{vs}(f|_J)$; (b)* $\mathrm{Err} \le (\ln 2)(L - \ell_\emptyset)$ *and* $L - \ell_\emptyset \le 2\,\mathrm{Err}\,(\mathrm{gad}(\lceil L \rceil)+3)$, *with* $\mathrm{gad}(j) := \mathrm{gg}\,(5 + 2\log(d+3j+\mathrm{gg}+1)) + \log(\mathrm{gg}+1)$ *and* $\mathrm{gg} := 2d+2$.

So Bayes over nondeterministic verifier circuits makes few posterior mistakes on f exactly when f has a small consistent verifier circuit, and, by Theorem A.43, exactly when f has a small finite axiomatisation in a fixed language whose data have polynomially short derivations, up to polynomial changes.

Proof. (a) The prior has an exponential tail in the size (the total weight of sizes above S is at most 2^{-S}), and the verifier with no active output disagrees with f at every size, so the argument of Theorem 9.8(a) of the companion paper applies; the bound on L is the weight of one smallest consistent verifier circuit for $f|_J$. (b) The first inequality is $\epsilon_i \le -\ln(1-\epsilon_i)$ with telescoping; now $W_{\mathrm{ver}}(\emptyset) < 1$, hence $\ell_\emptyset$. For the second we use an exception gadget for verifiers: given $\mathcal{V}$, a point a and a label b, append a circuit for $[x = a]$ (at most $2d-1$ gates), its negation, and new outputs $\mathsf{a}' := \mathsf{a} \vee [x=a]$, $\mathsf{r}' := \mathsf{r} \wedge \neg[x=a]$ if $b=1$ (symmetrically if $b=0$): at most $\mathrm{gg} = 2d+2$ gates. The new circuit $\mathcal{V}'$ is consistent (at a only the b-output can fire, and elsewhere it is $\mathcal{V}$), is defined at a with label b, agrees with $\mathcal{V}$ off a, and $\mathcal{V} \mapsto \mathcal{V}'$ is injective, with $w_{\mathrm{ver}}(\mathcal{V}')/w_{\mathrm{ver}}(\mathcal{V}) \ge 2^{-\mathrm{gad}(|\mathcal{V}|)}$. It maps the hypotheses of size at most S that are wrong or undefined at x_i injectively to hypotheses

that are right at x_i and still compatible with D_{i-1}, whose inputs differ from x_i. The rest of the proof of Theorem 9.8(b) of the companion paper is unchanged. □

Remark A.45 (The axiom-set prior itself). *Proof sketch.* We do not write out the converse direction for a prior over finite axiom sets; the obstacles are listed here. For a mixture over finite axiom sets with a derivation budget, the upper bounds $L = O(a \log a)$, with a the size of a smallest compatible axiom set, and Err $\leq (\ln 2)L$ hold directly (Theorem A.43(i) and telescoping). The converse needs an exception gadget for *every* hypothesis T: replace R by a fresh unary predicate R' in T and add $\forall x\,(x \neq t_a \to (R(x) \leftrightarrow R'(x)))$ and $R(t_a)^b$. This works if the language supplies fresh predicate symbols (so it is not a fixed finite language, and the symbol's index costs $O(\log)$ bits), if B contains the free-algebra axioms needed to derive $t_x \neq t_a$ (of size $O(d^2)$), and if the budget is charged in the prior, so that the gadget's extra derivation size $O(d^2)$ moves a hypothesis to a nearby budget at a cost of $O(\log)$ bits.

References

Heinrich Behmann. Beiträge zur Algebra der Logik, insbesondere zum Entscheidungsproblem. *Mathematische Annalen*, 86:163–229, 1922.

L. Berman. The complexity of logical theories. *Theoretical Computer Science*, 11(1):71–77, 1980.

George S. Boolos, John P. Burgess, and Richard C. Jeffrey. *Computability and Logic*. Cambridge University Press, Cambridge, fifth edition, 2007.

A. K. Chandra, D. C. Kozen, and L. J. Stockmeyer. Alternation. *Journal of the ACM*, 28(1): 114–133, 1981.

Alonzo Church. A note on the Entscheidungsproblem. *Journal of Symbolic Logic*, 1(1):40–41, 1936.

Stephen A. Cook and Robert A. Reckhow. The relative efficiency of propositional proof systems. *Journal of Symbolic Logic*, 44(1):36–50, 1979.

Martin Davis. Hilbert's tenth problem is unsolvable. *The American Mathematical Monthly*, 80 (3):233–269, 1973.

K. de Leeuw, E. F. Moore, C. E. Shannon, and N. Shapiro. Computability by probabilistic machines. In C. E. Shannon and J. McCarthy, editors, *Automata Studies*, volume 34 of *Annals of Mathematics Studies*, pages 183–212. Princeton University Press, Princeton, NJ, 1956.

Michael J. Fischer and Michael O. Rabin. Super-exponential complexity of Presburger arithmetic. In *Complexity of Computation*, volume 7 of *SIAM-AMS Proceedings*, pages 27–41. American Mathematical Society, Providence, RI, 1974.

Kaarel Hänni. Working notes, 2023–2026. Public repository, `https://github.com/kaarelh/notes`, 2026. Accessed October 2026. The note on Solomonoff axiom induction is in the folder `ai/learning/solomonoff induction`.

Kaarel Hänni and Claude (Anthropic). Bayesian axiom induction from instances: Template priors, derivation likelihoods and time penalties. Report, 2026a. Folder `axiom-induction` of `https://github.com/kaarelh/AI-works`.

Kaarel Hänni and Claude (Anthropic). Solomonoff operator induction: Definitions, versions and basic theorems. Preprint, 2026b. Folder `solomonoff-operator-induction` of `https://github.com/kaarelh/AI-works`.

Kaarel Hänni and Claude (Anthropic). Steeper simplicity priors in Solomonoff operator induction: The model–randomness split and the teacher's mistakes. Preprint, 2026c. Folder `solomonoff-operator-induction` of `https://github.com/kaarelh/AI-works`.

Juris Hartmanis and Richard E. Stearns. On the computational complexity of algorithms. *Transactions of the American Mathematical Society*, 117:285–306, 1965. doi: 10.1090/S0002-9947-1965-0170805-7.

Clemens Lautemann. BPP and the polynomial hierarchy. *Information Processing Letters*, 17(4): 215–217, 1983. doi: 10.1016/0020-0190(83)90044-3.

Leopold Löwenheim. Über Möglichkeiten im Relativkalkül. *Mathematische Annalen*, 76:447–470, 1915. doi: 10.1007/BF01458217.

Yu. V. Matiyasevich. Enumerable sets are Diophantine. *Soviet Mathematics Doklady*, 11(2): 354–358, 1970.

Nicholas Pippenger and Michael J. Fischer. Relations among complexity measures. *Journal of the ACM*, 26(2):361–381, 1979. doi: 10.1145/322123.322138.

Mojżesz Presburger. Über die Vollständigkeit eines gewissen Systems der Arithmetik ganzer Zahlen, in welchem die Addition als einzige Operation hervortritt. In *Comptes Rendus du Premier Congrès des Mathématiciens des Pays Slaves, Warszawa 1929*, pages 92–101, 1929.

Joel I. Seiferas, Michael J. Fischer, and Albert R. Meyer. Separating nondeterministic time complexity classes. *Journal of the ACM*, 25(1):146–167, 1978.

Michael Sipser. A complexity theoretic approach to randomness. In *Proceedings of the Fifteenth Annual ACM Symposium on Theory of Computing (STOC 1983)*. ACM, 1983.

Raymond M. Smullyan. *Theory of Formal Systems*. Number 47 in Annals of Mathematics Studies. Princeton University Press, Princeton, NJ, 1961.